

Errata to *Counting elliptic curves with a cyclic m -isogeny*

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This separata records the changes made in the corrected version of my thesis. Page references are to the physical pages of the submitted thesis PDF, counted from the title page as p. 1; thus thesis p. 1 is Submitted Thesis p. 9.

Submitted Thesis p. 3

-... power saving ...
+... power-saving ...

Submitted Thesis p. 4

-... blessed with support and ...
+... blessed with the support and ...

-... go out to to my ...
+... go out to my ...

Submitted Thesis p. 5

-... fellow-travelers who ...
+... fellow-travelers whom ...

-... experience has ...
+... experience have ...

Submitted Thesis p. 9

-... elliptic ...
+... elliptic ...

Submitted Thesis p. 13

-... \textup{unsigned} ...
+... \textup{an unsigned} ...

Submitted Thesis p. 15

-... \consteqprm $m X^{1/6} + O(X^{1/8 + \epsilon})$
+... \consteqprm $m X^{1/6} + O(X^{38/291 + \epsilon})$

-... \constadprm $m X^{1/6} + O(X^{1/8 + \epsilon})$
+... \constadprm $m X^{1/6} + O(X^{38/291 + \epsilon})$

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-\$\twerror m\$ & \$2\$ & \$6\$ & \$12\$ & \$12\$ & \$12\$ & \$12\$ & \$21\$ & \$24\$ & \$15\$ & \$24\$ & \$36\$ & \$33\$ \\
+\$\twerror m\$ & \$2\$ & \$6\$ & \$12\$ & \$12\$ & \$12\$ & \$12\$ & \$18\$ & \$24\$ & \$51/4\$ & \$24\$ & \$36\$ & \$30\$ \\

-... 7}, \Cref{Theorem: asymptotic for NQ(X) for m = ...
+... 7}, \Cref{Theorem: asymptotic for twN(X) for m = ...

-... and 25}, \Cref{Theorem: asymptotic for NQ(X) for m = 13}, and ...
+... and 25}, \Cref{Theorem: asymptotic for twN(X) for m = 13}, and ...

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-... $m(X)$ associated ...
+... $m(s)$ associated ...

Submitted Thesis p. 21

-... 5, where our estimates ...
+... 5, where we use our estimates ...

-... and we able to ...
+... and we are able to ...

Submitted Thesis p. 22

-... summarize the contents in the remainder of ...
+... summarize the remainder of ...

Submitted Thesis p. 23

-... $(X)_{\dots}$
+... $(X)_{\dots}$

Submitted Thesis p. 26

-... millenium ...
+... millennium ...

Submitted Thesis p. 27

- $E^{\{u^2\}} : y^2 : x^3 + u^4 \text{ A } \dots$
+ $E^{\{u^2\}} : y^2 = x^3 + u^4 \text{ A } \dots$

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-... = $E^{\{u^2\}} : y^2 : x^3 + u^4 \dots$
+... = $E^{\{u^2\}} : y^2 = x^3 + u^4 \dots$

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-... torsion ([\Cref{Theorem: Mordell's Theorem}](#) and [\Cref{Theorem: Mazur's Theorem}](#)). ...
+... torsion ([\Cref{Theorem: Mazur's Theorem}](#)). ...

-... $(x, -y)_{\dots}$; ...
+... $(x_0, -y_0)_{\dots}$; ...

-... passes though ...
+... passes through ...

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-... `groupplot{\includegraphics[scale=0.3]{Graphics/y2=x3-3x+62.png}}`{The ...
+... `groupplot{\includegraphics[scale=0.3]{Graphics/y^2=x^3-3x+62.png}}`{The ...

-... via its the isomorphism with ...
+... via its isomorphism with ...

Submitted Thesis p. 32

-... of $E_{\dots}$, we ...
+... of $E_{1\dots}$, we ...

Submitted Thesis p. 33

-... already gives ...
+... already give ...

-... $u^4 \text{ A } B = \dots$
+... $u^4 \text{ A } B$ and $B = \dots$

-... $E_2 \rightarrow E_2$ with
+... $E_2 \rightarrow E_1$ with

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-... $\subseteq E(\mathbb{Q})$ for ...
+... $\subseteq E_1(\mathbb{Q})$ for ...
-... out that the $\ker \phi$...
+... out that $\ker \phi$...

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- $f^{(c)}(t) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - x(P)) = \prod_{P \in \Phi} (t - c^2 x(P)) \in \mathbb{Q}[t]$
+ $f^{(c)}(t) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - x(P)) = \prod_{\substack{P \in \Phi \\ P \neq \infty}} (t - c^2 x(P)) \in \mathbb{Q}[t]$

Submitted Thesis p. 37

-... and only Φ is ...
+... and only if Φ is ...
-... E or ...
+... E of ...
-... 2-isogeny ...
+... 2 -isogeny ...

Submitted Thesis p. 38

-... order 25, ...
+... order p^2 , ...

Submitted Thesis p. 39

-... be cyclic n -isogenies ...
+... be a cyclic n -isogeny ...
-... There exist (infinitely many) elliptic curves ...
+... There exist elliptic curves ...

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-By \Cref{Proposition: factoring cyclic m-isogenies}, we can understand cyclic m -isogenies as compositions of cyclic ... is precluded by the absence of any isogeny graphs with more than two 5-isogenies.
+ By \Cref{Proposition: factoring cyclic m-isogenies} and \Cref{Theorem: Kenku-Mazer Theorem}, it suffices to consider the following levels. Namely, $m \in \{2,3,4,5,6,8,9,10,12,15,16,18,25,27\}$. The cases $m \in \{2,4,8,16\}$, $m=3$, $m=5$, and $m \in \{6,12\}$ follow by inspection of the graphs T_8 , $L_3(9)$, $L_3(25)$, and SS , respectively \cite[Tables 1,2,4]{Chiloyan-Lozano-Robledo}. For $m=9,27$, the classified graphs contain at most one cyclic subgroup of the relevant order from any vertex. For $m=10,15$, the rectangular graph $R_4(m)$ has unique rational primary components, hence at most one cyclic m -kernel. For $m=18$, the middle vertices of $R_6(18)$ have no cyclic 9 -isogeny, while at a corner the unique 9 - and 2 -primary kernels determine the 18 -kernel. Finally, $L_3(25)$ gives at most one cyclic 25 -kernel \cite[Tables 1,3]{Chiloyan-Lozano-Robledo}.

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-... 12, 16, 18, see ...
+... 12, 16, see ...

Submitted Thesis p. 42

-... 2.428\,650\,6\dots
+... 1.149\,249\,826\dots

Submitted Thesis p. 43

-... for an elliptic curve E (or more generally, of ...
+... of an elliptic curve E (or, more generally, ...

- $\alpha_{\text{hht}_F} \approx 349,068$
+ $\alpha_{\text{hht}_F} \approx 348,716$

Submitted Thesis p. 44

-... power-savings ...
+... power-saving ...

-... the embedding the coordinates ...
+... the embedding of the coordinates ...

-... this paper, when ...
+... this thesis, when ...

Submitted Thesis p. 45

-... this paper, we ...
+... this thesis, we ...

Submitted Thesis p. 46

-... with power-saving asymptotic for ...
+... with power-saving error for ...

Submitted Thesis p. 47

-... cyclic isogenies by ...
+... cyclic isogeny by ...

-... $m(X) \stackrel{\text{Equation: counts}}{\sim}$...
+... $m(X)$ from $\stackrel{\text{Equation: counts}}{\sim}$...

-... that $\not\sim$...
+... that $\not\sim$...

-... that $\sim$...
+... that $\sim$...

-The cases $m = \dots$
+The case $m = \dots$

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-... They parameterize ...
+... They parameterized ...

Submitted Thesis p. 49

-... well-behaved functions ...
+... well-behaved function ...

Submitted Thesis p. 50

-... an effective ...
+... an effectively ...

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```

-\begin{theorem}[\cite[Theorem 1.3]{Pizzo-Pomerance-Voight}]\label{Theorem: asymptotics for N(X) = 3}
-   There exist effective computable constants
+\begin{theorem}[\cite[Theorem 1.3, Proposition 2.9]{Pizzo-Pomerance-Voight}]\label{Theorem: asymptotics for N(X) = 3}
+   There exist effectively computable constants

-... 1.742\,501\,704\,06\ldots, ...
+... 1.671\,470\,808\,106\ldots, ...

-... -0.835\,735\,404\,05\ldots
+... -0.871\,250\,852\,030\ldots

```

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```

-... moduler ...
+... modular ...

```

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```

-... = 108$ while ...
+... = 1728$ while ...

-... 1458) ...
+... 5832) ...

```

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```

-y^2=x^3+(A/e^4)x\pm(B/e^3)
+y^2=x^3+(A/e^2)x\pm(B/e^3)

-... only models for ...
+... only twist minimal models in the quadratic twist class of ...

-... \eqref{Example: ...
+... \Cref{Example: ...

```

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```

-... \set{4, 5, 6, 8, 9, 10, ...
+... \set{4, 5, 6, 7, 8, 9, 10, ...

```

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```

-... $\mathbb{Q}$ are precisely ...
+... $\mathbb{Q}$ is precisely the set of ...

```

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```

-... $E$. If $E$ has ...
+... $E$. If $E_{\text{pr}}$ has ...

```

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```

-... equipped with cyclic 4-isogeny ...
+... equipped with a cyclic 4-isogeny ...

-... 1$ \set{t^4 + 268 t^3} ...
+... 1$ \set{t^4 + 268 t^3} ...

```

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```

-... & $\set{1, 2, \infty}$ ...
+... & $\set{1, 3/2, 2, \infty}$ ...

-... only unique up to ...
+... unique only up to a ...

```

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```
-... 27 B(a, ...
+... 27 \B m(a, ...
```

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```
-... 7(a,b),c^3\B 7(a,b))$.
+... m(a,b),c^3\B m(a,b))$.

-... minimality defect \eqref{Equation: defect powers}. ...
+... minimality defect defined in \eqref{Equation: twist defect powers}. ...
```

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```
-... with a  $m$ -isogeny ...
+... with an  $m$ -isogeny ...

-... parameterized ...
+... parameterized ...

-... 4, 5, 6, 7, 8, 9, 10, 12, 13, 16, 18, 25}.
+... 4, 5, 6, 7, 8, 9, 10, 12, 13, 16, 18, 25}
```

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```
-... cyclic  $n$ -isogenies whose ...
+... cyclic  $m$ -isogenies whose ...
```

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```
-...  $\Phi_2$  in  $\Phi_1$ , ...
+...  $\Phi_2$ , ...

-...  $n > 1$  ...
+...  $n < m$  ...
```

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```
-... and  $Q$  be generators ...
+... and  $Q$  to be generators ...

-...  $aQ$  generates  $\Phi_2$ , ...
+...  $a$  is a unit modulo  $m$ , ...

-... with  $aQ$  ...
+... with  $a^{-1}Q$  ...

-...  $\mathbb{Z} \subseteq \dots$ 
+...  $\mathbb{Z} \not\subseteq \dots$ 

-...  $\mathbb{Z}$ , then the  $\#\text{parent}\{\Phi_1 \dots$ 
+...  $\mathbb{Z}$ , then  $\#\text{parent}\{\Phi_1 \dots$ 
```

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```
- The image of  $\text{Gal}_{\mathbb{Q}}$  in  $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$  maps each of these vectors to a multiple of itself, and a little linear algebra shows that the set of all such matrices in  $M_2(\mathbb{Z}/m\mathbb{Z})$  is precisely the set
+ The image of  $\text{Gal}_{\mathbb{Q}}$  in  $\text{GL}_2(\mathbb{Z}/m\mathbb{Z})$  preserves each of the cyclic subgroups generated by these two vectors, and hence is contained in the subgroup

- \set{\begin{pmatrix}
- a & b \\ 0 & a + n b + n\text{prm } c
- \end{pmatrix} : a \in (\mathbb{Z}/m\mathbb{Z})^\times \text{ and } b, c \in \mathbb{Z}/m\mathbb{Z}}.
+ H_{\{m,n\}} \colonequals \set{\begin{pmatrix}
+ a & b \\ 0 & d
+ \end{pmatrix} \in \text{GL}_2(\mathbb{Z}/m\mathbb{Z}) : d \equiv a + n b \pmod{n\text{prm}}}.

- Conversely, if the image of  $\text{Gal}_{\mathbb{Q}}$  on  $\text{GL}(E[m](\overline{\mathbb{Q}}))$  is of the form given by \eqref{Equation: Galois image in GLm} for some choice of basis  $P_1, P_2$ , then  $E$  has two unsigned cyclic  $m$ -isogenies with kernels  $\Phi_1$  and  $\Phi_2$  having intersection of size  $n$ .
```

+ Conversely, if the image of $\text{Gal}_{\mathbb{Q}}$ on $E[m](\mathbb{Q})$ is contained in $H_{m,n}$ for some choice of basis P_1, P_2 , then the cyclic subgroups generated by P_1 and P_1+nP_2 are Galois-stable and have intersection of order n . Thus E has two unsigned cyclic m -isogenies with kernels having intersection of size n .

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-... $\mathbb{Q}$, and do not have any $\mathbb{Q}$ -points. ...
+... $\mathbb{Q}$

-... map from image of ...
+... map from the image of ...

-... image of $\text{Gal}_{\mathbb{Q}}$...
+... image of $H_{m,n}$ from $\text{Gal}_{\mathbb{Q}}$...

-... of $(\mathbb{Z}/m\mathbb{Z})^\times$. Thus, for ...
+... of $(\mathbb{Z}/m\mathbb{Z})^\times$. In particular, there are no elliptic curves $E/\mathbb{Q}$ possessing such a pair. Thus, for ...

-... of $\text{Chiloyan-Lozano-Robledo}$. '
+... of $\text{Chiloyan-Lozano-Robledo}$.

-... $\{12, 16, 18\}$. Then ...
+... $\{12, 16\}$. Then ...

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-... the moduli space for the space of ...
+... the moduli space of ...

-... -3 (3 ...
+... -3 (3 ...

-... n (t) for m , ...
+... n (t) for m , ...

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-... weaker conditions $\mathcal{R}$ than ...
+... weaker conditions on $\mathcal{R}$ than ...

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- $\text{Area}(\text{calR}_m(X)) \neq \text{Area}(\{(X^{1/2 \deg B} a, X^{1/2 \deg B} b) : (a, b) \in \text{calR}_m(1)\})$
+ $\text{Area}(\text{calR}_m(X)) \neq \text{Area}(\{(X^{1/2 \deg B} a, X^{1/2 \deg B} b) : (a, b) \in \text{calR}_m(1)\})$

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-... to $O(X^{1/2 \deg B})$...
+... to $O(X^{1/2 \deg B})$...

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-... -3 , ...
+... $-3b^2$, ...

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-... $\mathcal{L}$, and independent of ...
+... $\mathcal{L}$, and is independent of ...

-... $\sqrt{2} = 4.828\dots$
+... $\sqrt{5} = 6.472\dots$

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```
-... Let  $\sigma, \rho$  in ...
+... Let  $\sigma$  in ...

-...  $t^{-\sigma}$  ...
+...  $u^{-\sigma}$  ...

-... Bingham--Glodie--Teugels ...
+... Bingham--Goldie--Teugels ...
```

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```
-...  $\frac{X}{\zeta(k)}$  ...
+...  $\frac{X}{\zeta(2)}$  ...
```

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```
-... and  $L_{\beta(s)}$  both converge absolutely ...
+... and  $L_{\beta(s)}$  converge absolutely ...
```

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```
-... be complex functions functions of ...
+... be complex functions of ...

- $\mu_{FG}(\sigma) = \dots$ 
+ $\mu_{FG}(\sigma) \leq \dots$ 
```

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```
-... to Roux \cite{Roux}
+... to Roux \cite{Roux}.

-...  $\delta/\xi$  ...
+...  $\delta/\lfloor \xi \rfloor$  ...
```

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```
-... the Euler-Mascheroni constant ...
+... the Euler--Mascheroni constant ...
```

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```
-for any  $\epsilon > 0$ , ...
+For any  $\epsilon > 0$ , ...

-... 0.607927  $X$  ...
+... 0.607927  $y$  ...

-... has a simple pole of ...
+... has a pole of ...

-...  $\mu_{\zeta}(\sigma) = \dots$ 
+...  $\mu_{\zeta}(\sigma) \leq \dots$ 
```

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```
-... for addressing third step ...
+... for addressing the third step ...
```

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```
-... that admits ...
+... that admit ...

-... by  $\text{twistNeq}$  ...
+... by  $\text{twNad}$  ...
```

-... are important difference between ...
+... are important differences between ...

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-... $\{\text{calE}\} (X) = \sum_{n \geq 1} \text{twNad} \dots$
+... $\{\text{calE}\} (X) = \sum_{e \geq 1} \text{twNad} \dots$

-... f), \label{Equation: ...
+... f). \label{Equation: ...

- and applying M^{obius} inversion to $\text{eqref{Equation: cM}(X e)}$ in terms of $\text{twN}(X e)$ yields
- \begin{equation}
- \text{twNad} \{\text{calE}\} (X; e) = \sum_{f \mid X^{1/6}} \mu(f) \text{cM} \text{calE} (e^6 f^6 X; e f).
- \end{equation}
- Note, however, that any $E \in \text{calE}$ with $H(E) > e^6 X$ cannot contribute to $\text{twNad} \{\text{calE}\} (X; e)$, so in fact we have the equality
+
+ For the inversion, we must keep the raw-height cutoff fixed. A model counted by $\text{twNad} \text{calE}(X; e)$ has $\text{tmd}(E) = e$ and $\text{rawheight}(E) \leq e^6 X$. Inclusion--exclusion among the multiples of e therefore gives

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- $\text{twNad} \{\text{calE}\} (X; e) = \sum_{f \mid X^{1/6}} \mu(f) \text{cM} \text{calE} (e^6 X; \dots$
+ $\text{twNad} \{\text{calE}\} (X; e) = \sum_{f \geq 1} \mu(f) \text{cM} \text{calE} (e^6 X; \dots$

-... e f). \label{Equation: ...
+... e f), \label{Equation: ...

-Substituting $\text{eqref{Equation: ...}$
+where the sum is finite. Substituting $\text{eqref{Equation: ...}$

-... a modest ...
+... an absolute-value ...

- For all $e, n \in \mathbb{Z}_{>0}$ and $X > 0$, if $e \mid n$ then we have
- \begin{equation}
- $0 \leq \sum_{e \mid n} \mu(n/e) \text{cM} \text{calE}(e^6 X; n) \leq \text{cM} \text{calE}(n^6 X; n)$.
- \end{equation}
+For all $n \geq 1$ and $X > 0$,
+ \begin{equation}
+ $|\sum_{e \mid n} \mu(n/e) \text{cM} \text{calE}(e^6 X; n)| \leq \sum_{e \mid n} |\mu(n/e)| \text{cM} \text{calE}(e^6 X; n)$.
+ \end{equation}
+Every model on the right has twist height at most X , and after summing over n a model of twist minimality defect D occurs with multiplicity at most $\tau(D)^2$.

-By $\text{eqref{Equation: defining cM}}$, $\text{cM} \text{calE} (X; e)$ counts elliptic curves $E \in \text{calE} \dots > e^6 X$ for all e a proper divisor of n . The claim follows.
+The displayed inequality is the triangle inequality. If E is counted with $e \mid n \mid D$ $\text{colonequals} \text{tmd}(E)$, then $\text{rawheight}(E) \leq e^6 X \leq D^6 X$; its total multiplicity is at most $\sum_{n \mid D} 2^{2\omega(n)} \leq \tau(D)^2$.
+ \end{proof}
+
+ \begin{lemma} \label{Lemma: weighted 2omega divisor factor}
+For fixed $\alpha > 0$, $\beta > 1$, and $\gamma \geq 2$,
+ \begin{equation}
+ $\sum_{n > y} \frac{2^{2\omega(n)}}{\prod_{p \mid n} (1 + p^{-\alpha})} n^{-\beta} = O_{\alpha, \beta} \left(\frac{\log y}{y^{\beta-1}} \right)$.
+ \end{equation}
+ \end{lemma}
+
+ \begin{proof}
+Since $\prod_{p \mid n} (1 + p^{-\alpha}) \leq \sum_{d \mid n} d^{-\alpha}$ and $2^{2\omega(dm)} \leq 2^{2\omega(d)} 2^{2\omega(m)}$, $\text{Cref{$
Corollary: sum of $2^{2\omega(n)}$ gives partial sums $O_{\alpha} (x \log x)$; partial summation proves the claim.

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-... elliptic curve in ...
+... elliptic curves in ...

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-... repetition of ...
+... repetition if ...

-... $E \in \text{scrE}$ of twist ...
+... $E \in \text{calE}$ of twist ...

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```
-... $\NQad \calE (s)$.
+... $\NQad \calE (X)$.

-... for $\LQad \calE (s)$. ...
+... for $\NQad \calE (X)$. ...

-... \twistLad \calE (n), \ \hQeq ...
+... \twistLad \calE (s), \ \hQeq ...

-... \hQeq m (n) &\colonequals \hQad \calE, \ \text{and} \ \
+... \hQeq m (n) &\colonequals \hQad \calE (n), \ \text{and} \ \
```

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```
-... $\hQeq ...
+... $\twistheq ...

-... \twistLad \calE (n), \ \hQad ...
+... \twistLad \calE (s), \ \hQad ...

-... \hQad m (n) &\colonequals \hQad \calE, \ \text{and} \ \
+... \hQad m (n) &\colonequals \hQad \calE (n), \ \text{and} \ \

-... number of of elliptic curves ...
+... number of elliptic curves ...
```

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```
-... prove \Cref{Intro Theorem: asymptotic for NQ(X) for 5 < m <= 9} when $m = 7$, with ...
+... prove \Cref{Intro Theorem: asymptotic for twN(X) for m of genus $0$} when $m = 7$, with ...
```

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```
-... discussssed ...
+... discussed ...

-... $7$-groomed, let $\ell$ be prime, and let $v \in \Z_{\geq 0}$. Then the following statements hold.
+... $7$-groomed, let $\ell$ be prime, and let $v \in \Z_{>0}$. Then the following statements hold.
```

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```
-... by \eqref{Equation: defect powers}, ...
+... by \eqref{Equation: twist defect powers}, ...

-so $\ell \nmid \gcd(\tB 7(a,b), \{C 7\}(a, b))$ and thus $\ell^v \mid \tmd(\A 7(a,b), \B 7(a,b))$ if and only if $\ell^{3v} \mid \mid{C 7}(a,b)$. If $\ell = 2$, a short computation confirms that $\B 7(a, b)$ is odd whenever $(a, b)$ is $7$-groomed, so our claim also holds in this case.
+so $\ell \nmid \gcd(\tB 7(a,b), \{C 7\}(a, b))$ and thus $\ell^v \mid \tmd(\A 7(a,b), \B 7(a,b))$ if and only if $\ell^{3v} \mid \mid{C 7}(a,b)$. If $\ell = 2$, then $\C 7(a,b)$ and $\tA 7(a,b)$ are both odd whenever $(a,b)$ is $7$-groomed, so $\A 7(a,b)$ is odd and our claim also holds in this case.

-... 7(b,0) ...
+... 7(a,0) ...

-... $\G$ to cancel all occurences of the variable $a$, ...
+... $\G$ to cancel all occurences of the variable $a$, ...
```

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```
-... $\G$ to cancel all occurences of the variable $b$, ...
+... $\G$ to cancel all occurences of the variable $b$, ...
```

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```
-... $(F, G) = \in \set{(\tA ...
+... $(F, G) \in \set{(\tA ...

-... -3\parent{13 ...
+... 3\parent{13 ...
```

```

-... -9\parent{303 ...
+... 9\parent{303 ...

-... 1763 a - ...
+... -1763 a + ...

-          \gcdP {\tB 7} {\C 7} 2 (a, b) &= -\parent{13 a + 20 b}, \ \text{and} \ \
+          \gcdP {\tB 7} {\C 7} 2 (a, b) &= 13 a + 20 b, \ \text{and} \ \

-... a - b), ...
+... a + b), ...

-... \twistdefect(A(a, b), B(a, ...
+... \twistdefect(\A 7(a, b), \B 7(a, ...

```

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```

-... Now it ...
+... Now if ...

-Now let  $p = 2$ . A short computation confirms that for  $a$  and  $b$  coprime,  $\{ \tB 7 \}(a, b) \not\equiv 0 \pmod{2}$  and  $\{ \C 7 \}(a, b) \not\equiv 0 \pmod{2}$ , so our claim holds vacuously in this case.
+Now let  $p = 2$ . For  $a$  and  $b$  coprime,  $\{ \tA 7 \}(a, b) \not\equiv 0 \pmod{2}$  and  $\{ \C 7 \}(a, b) \not\equiv 0 \pmod{2}$ , so  $\{ \A 7 \}(a, b)$  is odd and our claim holds vacuously in this case.

-...  $b$  and cube of ...
+...  $b$  and the cube of ...

```

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```

-... \mid g 7(t) ...
+... \mid f 7(t) ...

```

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```

-\item For all prime  $\ell \neq 3, 7$  ...
+\item For every prime  $\ell \neq 3, 7$  ...

```

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```

-Hensel's lemma still applies to  $\h 7(t)$ : let  $t_0, t_1$  be the roots of  $\h 7(t)$  in  $\Z_7$  with  $t_0 \colonequals 248\,044 \pmod{7^7}$  (so that  $t_1 = -1 - t_0$ ). We claim that
+Hensel's lemma still applies to  $\h 7(t)$ : let  $t_0, t_1$  be the roots of  $\h 7(t)$  in  $\Z_7$  with  $t_0 \colonequals 248\,044 \pmod{7^7}$  (so that  $t_1 = -1 - t_0$ ). The branch  $t_0 \equiv -1 \pmod{7}$  has  $\tg 7(t_0) \not\equiv 0 \pmod{7}$ , while on the branch  $t_1 \equiv 0 \pmod{7}$  one checks that  $7^4 \mid \tf 7(t)$  and  $7^7 \nmid \tg 7(t)$  whenever  $t \equiv 42 \pmod{49}$ . Consequently, for  $v \geq 3$  we have

-\calT 7(7^{3v}) ...
+\calT 7(7^v) ...

-for  $3v \geq 7$ . Indeed,  $\tg 7(t_1) \equiv 0 \pmod{7^7}$ , so we can afford to approximate  $t_1$  modulo  $7^{3v-7}$ . As  $g 7(t_0) \not\equiv 0 \pmod{7}$  and  $g 7(t_1) \not\equiv 0 \pmod{7^8}$ , no other values of  $t$  suffice. Thus  $\calT 7(7^{3v}) = 1 + 7^7 = 823544$ .
+because the first branch requires  $7^{3v} \mid \h 7(t)$ , whereas on the second branch  $7^{3v-7} \mid \h 7(t)$  is necessary and sufficient for the two divisibility conditions defining  $\calT 7(7^v)$ . Thus  $\calT 7(7^v) = 1 + 7^7 = 823544$ .

```

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```

-}\{\calT 7(7^v) for  $v \geq 2$ 
+}\{\calT 7(7^v) for  $v = 1, 2$ 

-... of  $\bbZ[\zeta_6]$  ...
+... of  $\bbZ[3 \zeta_6]$  ...

```

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```

-...  $\bbZ[\zeta_6]$ . ...
+...  $\bbZ[3 \zeta_6]$ . ...

-...  $\zeta_6$ ,  $e$  ...
+...  $\zeta_6$ ,  $e^k$  ...

```

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```
-... $p$ prime ...  
+... a prime $p$ ...
```

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```
-... defect for m = 7}(c),  
+... defect for m = 7}(b),  
  
-e_0^{12} m^6 ...  
+e_0^{12} n_0^6 ...  
  
-... rounding $m$ ...  
+... rounding $n_0$ ...  
  
-... \cdot 7^7$ ...  
+... \cdot 7^3$ ...
```

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```
-... attained when $t = 0$ ...  
+... attained when $\theta = 0$ ...  
  
-... attained when $t = 4.980\,802\,4\ldots$ ...  
+... attained when $\theta = 4.980\,802\,4\ldots$ ...  
  
-... -0.371\ldots)$$. ...  
+... -0.371\ldots)$$. ...  
  
-... 1))\{C(a, ...  
+... 1))\{C 7(a, ...
```

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```
-... size of order of growth for ...  
+... order of growth of ...  
  
-for $X \geq 2$, $d$, $e \geq 1$.  
+for $X \geq 2$, $e \geq 1$.
```

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```
-... {e^3}, (a, b) ...  
+... {e^3}, (a:b) ...  
  
-... L_7(X; d, ...  
+... L_7(X; e, ...
```

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```
-... 7(x; ...  
+... 7(X; ...  
  
-\cM 7(x;e)  
+\cM 7(X;e)
```

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```
-... 7 (c\prm, d\prm) \ ...  
+... 7 (c\prm, d\prm)) \ ...
```

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```
-... \pm b_0 ...  
+... \pm \abs{b_0} ...  
  
-... b_0)) \leq \parent{\frac{101 + 16 \sqrt{37}}{27}}^{1/2} e_0^{1/2}.  
+... b_0)) \leq e_0^{1/2}.
```

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- These extrema are both attained when $a_0/b_0 = -1/2$.
- + The lower extremum is attained when $a_0/b_0 = -1/2$, and the upper extremum is attained when $b_0 = 0$.

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- ... $\displaystyle{\frac{2}{p^2 + 1}}$, ...
- +... $\displaystyle{\frac{2}{(p+1)^2}}$, ...
- ... $\frac{2}{p^2 + 1}$
- +... $\frac{2}{(p+1)^2}$

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- By \Cref{Lemma: bound on T(e) for m = 7}(f), $\mathcal{T}_m(e) = O(2^{-\omega(e)})$, so by \Cref{Lemma: asymptotic for M(X; e) for m = 7}, we have
- + By \Cref{Lemma: bound on T(e) for m = 7}(f), \Cref{Lemma: asymptotic for M(X; e) for m = 7}, and \Cref{Proposition: bounding the summands of twN calE X},
- $\mathcal{M}_m(X; e) = O(\text{parent}\{\frac{2^{-\omega(e)}}{X^{1/6}}\}e^3)$.
- + $\text{abs}\{\text{twistNadgy } 7(X)\} \ll X^{1/6} \sum_{n>y} \frac{2^{-\omega(n)}}{n^2} \prod_{p \mid n} (1 + \frac{1}{p})$.
- Now by \Cref{Proposition: bounding the summands of twN calE X}, we see \begin{equation} \text{twistNegy} \dots \text{ of } f/n^\sigma, \text{ we conclude } \begin{equation} \text{twistNegy } m(X) = O(\text{parent}\{\frac{X^{1/6}}{\log y}\}) \end{equation} as desired.
- + The result follows from \Cref{Lemma: weighted 2omega divisor factor} with $\alpha=1$ and $\beta=2$.

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- $\mu_{\text{twistLad } 7}(\sigma) < 13/84$
- + $\mu_{\text{twistLad } 7}(\sigma) \leq \max\{0, 2-12\sigma\}$

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- Finally, we prove part (c). By \Cref{Theorem: absolutely convergent Dirichlet series have mu = 0}, $\mu_{\text{twistLadR } 7}(\sigma) = 0$ for $\sigma > 1/12$, so by \Cref{Proposition: mu acts like a negative valuation} and \Cref{Theorem: muzeta(sigma)},
- + Finally, we prove part (c). For fixed $1/12 < \sigma_0 < 1/6$, \eqref{equation: bound on twLR for m = 7} gives $\mu_{\text{twistLadR } 7}(\sigma_0) \leq 1$, while absolute convergence gives $\mu_{\text{twistLadR } 7}(\sigma) = 0$ for $\sigma > 1/6$. By subconvexity of μ \cite[Theorem II.1.20]{Tenenbaum},
- $\mu_{\text{twistLad } 7}(\sigma) = \mu_{\text{zeta}_6}(\sigma) < 13/84$
- + $\mu_{\text{twistLadR } 7}(\sigma) \leq 2-12\sigma \ll (1/12 < \sigma < 1/6)$.
- for $\sigma > 1/12$. Our claim follows.
- + Since \Cref{Theorem: muzeta(sigma)} gives $\mu_{\text{zeta}_6}(\sigma) \leq \frac{13}{42}(1-6\sigma) < 2-12\sigma$ on the same strip, \eqref{equation: Defining twLR for m = 7} and \Cref{Proposition: mu acts like a negative valuation} yield $\mu_{\text{twistLad } 7}(\sigma) \leq \max\{0, 2-12\sigma\}$ for $\sigma > 1/12$.

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- ... part (a), since $\zeta(s)$ is nonvanishing when $\sigma > 1$, the ratio $\zeta(6s)/\zeta(12s)$ is meromorphic function for $\sigma > 1/12$. But \Cref{Corollary: twL(s) has a meromorphic continuation} ...
- +... part (a), since $\zeta(s)$ is nonvanishing when $\sigma > 1$, the ratio $\zeta(6s)/\zeta(12s)$ is a meromorphic function for $\sigma > 1/12$. But \Cref{Corollary: twL(s) has a meromorphic continuation} ...

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- The sequence $\text{parent}\{\text{hQad } 7(n)\}_{n \geq 1}$ is admissible \textup{(\Cref{Definition: admissible sequences})} with parameters $(1/6, 1/12, 13/42)$.
- + For every δ with $0 < \delta < 7/97$, the sequence $\text{parent}\{\text{hQad } 7(n)\}_{n \geq 1}$ is admissible \textup{(\Cref{Definition: admissible sequences})} with parameters
- + \begin{equation}
- + $\text{parent}\{1/6, \delta, 97\delta/7\}$.
- + \end{equation}
- ... σ_a , so ...
- +... $> 1/6$, so ...

-For (ii), \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7} tells us that $\text{twistLad } 7(s)$ has a meromorphic continuation when $\sigma = \text{repart}(s) > 1/12$; on the other hand, as $\zeta(12s)$ is nonvanishing for $\sigma > 1/12$, we see that $\zeta(6s)/\zeta(12s)$ has a meromorphic continuation to $\sigma > 1/12$, and so (ii) holds with

+For (ii), \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7} tells us that $\text{twistLad } 7(s)$ has a meromorphic continuation when $\sigma = \text{repart}(s) > 1/12$; on the other hand, as $\zeta(12s)$ is nonvanishing for $\sigma > 1/12$, we see that $\zeta(6s)/\zeta(12s)$ has a meromorphic continuation to $\sigma > 1/12$. Since $\delta < 7/97 < 1/12$, condition (ii) holds on the smaller half-plane $\sigma > 1/6 - \delta$. (The only pole of $L_{\text{Qad } 7}(s)/s$ there is the double pole at $s = 1/6$ indicated in \Cref{Theorem: relationship between twistL(s) and L(s) for m = 7}(b).)

+
+For (iii), suppose first that $1/6 - \delta < \sigma \leq 1/6$, and put $d = 1/6 - \sigma$, so $0 \leq d < \delta$. By \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7},

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$-\delta = 1/6 - 1/12 = 1/12$.
+ $\mu_{\text{twistLad } 7}(\sigma) \leq 12d$.

-(The only pole of $L_{\text{Qad } 7}(s)/s$ with $\sigma > 1/12$ is the double pole ... m = 7), $\mu_{\text{twistLad } 7}(\sigma) < 13/84$. Let $\zeta_a(s) = \zeta(as)$. Applying \Cref{Theorem: muzeta(sigma)}, we have
+Writing $\zeta_a(s) = \zeta(as)$, \Cref{Theorem: muzeta(sigma)} gives

$-\mu_{\zeta_6}(\sigma) = \mu_{\zeta_6}(\sigma) < \frac{13}{42} \left(1 - \frac{6}{12}\right) = \frac{13}{84}$ \end{equation} if $\sigma \leq 1/6$, ... the sequence $\left\{ \frac{1}{n} \log L_{\text{Qad } 7}(n) \right\}_{n \geq 1}$ is admissible with final parameter $\xi = 13/42$.
+ $\mu_{\zeta_6}(\sigma) = \mu_{\zeta_6}(\sigma) \leq \frac{13}{42} (1 - 6\sigma) = \frac{13}{7}d$,
+\end{equation}
+and, as $\zeta(12s)^{-1}$ is absolutely convergent for $\sigma > 1/12$, we have $\mu_{\zeta_{12}^{-1}}(\sigma) = 0$. Therefore \Cref{Theorem: relationship between twistL(s) and L(s)}(b) and \Cref{Proposition: mu acts like a negative valuation} imply
+\begin{equation}
+ $\mu_{L_{\text{Qad } 7}}(\sigma) \leq \frac{13}{7}d = \frac{97}{7}d < \frac{97}{7}\delta$.
+\end{equation}
+For $\sigma > 1/6$, all three factors are absolutely convergent, so $\mu_{L_{\text{Qad } 7}}(\sigma) = 0$. Thus condition (iii) holds with $\xi = 97\delta/7$, proving the claim.

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... $X^{1/6} + O(\text{parent}\{X^{1/8 + \epsilon}\})$
+... $X^{1/6} + O(\text{parent}\{X^{38/291 + \epsilon}\})$

- By \Cref{Lemma: Delta NQ(n) is admissible for m = 7}, the sequence $\left\{ \frac{1}{n} \log L_{\text{Qad } 7}(n) \right\}_{n \geq 1}$ is admissible with parameters $\{1/6, 1/12, 13/42\}$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series $L_{\text{Qad } 7}(s)$, and our claim follows.
+ Apply \Cref{Theorem: Landau's Tauberian theorem} and let $\delta \rightarrow 7/97$ in \Cref{Lemma: Delta NQ(n) is admissible for m = 7}; the exponent tends to $1/6 - 7/194 = 38/291$.

-We believe that with sufficient care and appropriate hypotheses, the denominator $\lfloor \xi \rfloor + 2$... for $L_{\text{NQad } 7}(X)$, like $\text{twNad } 7(X)$, would be reduced to $O(X^{1/12 + \epsilon})$.
+ The loss from the twist-counting error $O(X^{1/12} \log^5 X)$ to the exponent $38/291$ above comes from the vertical growth of the analytically continued remainder in \eqref{equation: twistLR(s) as an integral for m = 7}. We believe an elementary argument applying Dirichlet's hyperbola method to \Cref{Theorem: relationship between twistL(s) and L(s)}(a) could achieve a comparable, and perhaps modestly improved, error term without this analytic-growth loss.

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... $(X/108)^{1/6}$ (this bound arises ...
+... $\text{parent}\{3^3 \cdot 7^{18} X^{1/6}\}$ (this bound follows ...

... 7(a)). ...
+... 7(b)). ...

... $\gcd(a, b) = 1$, $b > 0$, and $\mathcal{C}_7(a, b)$...
+... $\gcd(a, b) = 1$, and $\mathcal{C}_7(a, b)$...

... $\gcd(a, b) = 1$, $b > 0$, and $\mathcal{C}_7(a, b)$...
+... $\gcd(a, b) = 1$, and $\mathcal{C}_7(a, b)$...

... m), $e_0^{12} m^6 \leq X/108$, ...
+... m) with $4e_0^{12} m^6 \leq 3^3 \cdot 7^{18} X$...

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... $b_e \text{parent}\{-1+3\zeta_6\}^3$...
+... $b_e \text{prmparent}\{-1+3\zeta_6\}^3$...

... by \ref{Lemma: ...
+... by \Cref{Lemma: ...

-... twice ...
+... twist ...

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-Reorganizing the sum in \Cref{Theorem: relationship} ...
+Reorganizing \Cref{Theorem: relationship} ...

-\NQuad 7 (X) = 2 \sum_{n \leq X} \sum_{c \leq (X/n)^{1/6}} \twistad 7 \parent{n/c^6} \abs{\mu(c)}.
+\NQuad 7 (X)=2\sum_{n \leq X} \twistad 7 (n) \, \kfree 2 \parent{(X/n)^{1/6}}.

-... in \Cref{Theorem: asymptotic for NQ(X) for m = 7} ...
+... in \Cref{Theorem: asymptotic for twN(X) for m = 7} ...

-... \leq Y \, \, p \equiv 1 \, \pmod 3 \, \, \parent{1 + \frac{2}{p^2+1}}
+... \leq Y \, \, p \equiv 1 \, \pmod 3 \, \, \parent{1 + \frac{2}{(p+1)^2}}

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-For \$a, b ...
+For \$e, a_0 ...

-... \, p \equiv a_0 ...
+... \, p \leq X, \, p \equiv a_0 ...

-In particular, letting \$Y = 10^{12}\$, we compute
+In particular, letting \$Y = 8 \cdot 10^9\$, we compute

-17.460\,405\,231\,126\,620 < \, Q \, 7 < 17.460\,405\,231\,134\,835
+17.460\,405\,236\,640\,390 < \, Q \, 7 < 17.460\,405\,236\,985\,885.

-This computation took approximately \$9\$ CPU days. % 789737 seconds.

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-We find \$r_7 \colonequals 243\,228\,665\,965\$ of our first \$s_7 \colonequals 595\,055\,000\,000\$ samples lie in \$\mathcal{R} \, 7\$, so
-\begin{equation}
-\mathcal{R} \, 7 \approx 0.105612 \cdot \frac{r_7}{s_7} = 0.04316889 \ldots
-\end{equation}
-with standard error
+We find \$r_7 \colonequals 243\,228\,665\,965\$ of our first \$s_7 \colonequals 595\,055\,000\,000\$ samples lie in \$\mathcal{R} \, 7\$, so

-0.105612 \cdot \sqrt{\frac{r_7(s_7 - r_7)}{s_7^3}} < 6.8 \cdot 10^{-8}.
+\mathcal{R} \, 7 \approx 0.105612 \cdot \frac{r_7}{s_7} = 0.043168893413 \ldots

-This took 11 CPU weeks to compute. Thus \$\twconst 7 = 0.45822276 \ldots\$, with error bounded by \$6.6 \cdot 10^{-7}\$\$.
+with standard error \$6.73053 \ldots \cdot 10^{-8}\$. This took 11 CPU weeks to compute. Using the midpoint of our interval for \$\mathcal{R} \, 7\$,
we obtain \$\twconst 7 \approx 0.4582228478 \ldots\$ with Monte Carlo standard error \$7.15 \cdot 10^{-7}\$\$.

-We have the identity \$\constad 7 = \twconstad 7 / 3 \, \zeta(2)\$, so \$\constad 7 = 0.092\,855\,36 \ldots\$ with an error of \$6.02 \cdot 10^{-8}\$.
+We have the identity \$\constad 7 = \twconstad 7 / 3 \, \zeta(2)\$, so \$\constad 7 \approx 0.092\,855\,36263 \ldots\$ with Monte Carlo
standard error \$1.45 \cdot 10^{-7}\$\$.

-We now turn our attention to \$\constadprm 7\$. As an intermediate step, we wish ... that for some \$M > 0\$ and for all \$u > X\$, we
have
+We now turn our attention to \$\constadprm 7\$. Truncating \eqref{Equation: Definition of ell0 for m = 7} at \$X=10^{42}\$ and using our
list of \$4\,582\,079\$ twist-minimal curves gives

-\abs{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} < M u^{1/12} \log^5 u.
+\ell_{7,0} \approx -0.463\,534\,394.

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-Thus
+In \Cref{Theorem: asymptotic for twN(X) for m = 7}, we have shown that for some \$M>0\$ and all \$u>X\$,
+\begin{equation}
+\abs{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} < M u^{1/12} \log^5 u.
+\end{equation}
+Thus the omitted tail in \$\ell_{7,0}\$ satisfies

-&\abs{\int_X^\infty \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \, \, \\-&\qquad \qquad < M \int_X^\infty u^{-13/12} \log^5 u \, \mathrm{d}u \, \, \\-&\qquad \qquad = 12 M X^{-1/12} (\log^5 X + 60 \log^4 X + 2880 \log^3 X \, \, \\-&\qquad \qquad + 103680 \log^2 X + 2488320 \log X + 29859840);
+&\abs{\frac{16}{\int_X^\infty \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, \mathrm{d}u} \, \, \\

+&\quad<2M $\times^{-1/12}$ (\log⁵X+60\log⁴X+2880\log³X\

+&\quad\quad+103680\log²X+2488320\log X+29859840).

-this gives us a bound on our truncation error.

-We do not know the exact value for M , but empirically, we find that for $1 \leq u \leq 10^{42}$, we have

+We do not know the exact value of M . For the large values of u examined in our computation up to 10^{42} , we find

--5.11 $\cdot 10^{-6} \leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12} \log^5 u} \leq 6.29 \cdot 10^{-7}$.

+5.11 $\cdot 10^{-6} \leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12} \log^5 u} \leq 6.29 \cdot 10^{-7}$.

-If we assume these bounds continue to hold for larger u , we find the truncation error for $\ell_7(0)$ is bounded by 68.95 , which catastrophically dwarfs our initial estimate.

+If we assume these inequalities continue to hold for $u > 10^{42}$, the resulting tail error is less than 56.73 .

-We can do better by sidestepping the logarithms. We know that $\text{twistNad } 7(X) - \text{twconst } 7 X^{1/6} = O(X^{1/12 + \epsilon})$ for every $\epsilon > 0$. We let $\epsilon = 10^{-4}$, and find that for $1 \leq u \leq 10^{42}$,

+We can do better by sidestepping the logarithms. Taking $\epsilon = 10^{-4}$, over the values examined up to 10^{42} we find

--1.2174 $\leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12 + \epsilon}} \leq 0.52272$.

+1.2174 $\leq \frac{\text{twistNad } 7(u) - \text{twconst } 7 \lfloor u^{1/6} \rfloor}{u^{1/12 + \epsilon}} \leq 0.52272$.

-If we assume these bounds continue to hold for larger u , we get an estimated truncation error of $2.43 \cdot 10^{-5}$, which is much more manageable.

+If we assume these inequalities continue to hold for larger u , the resulting tail error is less than $7.79 \cdot 10^{-4}$. Propagating the Monte Carlo standard error in $\text{twconstad } 7$ through the truncated integral contributes $1.09 \cdot 10^{-5}$, giving reported uncertainties 56.73 or $7.90 \cdot 10^{-4}$, respectively.

-Our estimate of $\ell_7(0)$ is also skewed by our estimates of $\text{twconstad } 7$. An error of ϵ in our estimate for $\text{twconstad } 7$ induces an error of

+Using [Equation: constants for m = 7](#), we obtain

- $\frac{\epsilon}{\log X} \int_1^X \frac{\lfloor u^{1/6} \rfloor}{u^{7/6}} du < \frac{\epsilon}{\log X} \int_1^X \frac{u^{-1}}{u} du = \frac{\epsilon}{\log X}$... asymptotic for $NQ(X)$ for $m = 7$. As a sanity check, we verify that

+ $\text{constadpr } 7 \approx -0.164, 049, 781$,

+ $\text{end{equation}}$

+with reported uncertainties 68.98 or $9.59 \cdot 10^{-4}$ under the two extrapolations above. As a sanity check,

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- $\frac{NQ(10^{42})}{10^7} - 42 \text{constad } 7 \log 10 = -0.164, 186, 667 \dots \approx \text{constadpr } 7$,

+ $\frac{NQ(10^{42})}{10^7} - 42 \text{constad } 7 \log 10 = -0.164, 192, 299 \dots \approx \text{constadpr } 7$,

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... Theorem: asymptotic for $twN(X)$ for m of nonzero genus ([Theorem: asymptotic for \$twN\(X\)\$ for \$m = \dots\$](#))

+... Theorem: asymptotic for $twN(X)$ for m of genus 0 ([Theorem: asymptotic for \$twN\(X\)\$ for \$m = \dots\$](#))

... forces us change how ...

+... forces us to change how ...

... (t). In [Section: Establishing notation when \$m = 10\$ and ...](#)

+... (t). In [Section: Estimating \$twN\(X\)\$ for \$m = 5\$ and \$10\$ and ...](#)

... to prove [Intro Theorem: asymptotic for \$NQ\(X\)\$ for \$m > 9\$](#) for $m = 10, 25$

+... to prove [Intro Theorem: asymptotic for \$twN\(X\)\$ for \$m\$ of genus \$0\$](#) for $m = 10, 25$

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... (a/b, ...

+... (a/b), ...

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... m -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{\geq 0}$. Then the following statements hold.

+... m -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{>0}$. Then the following statements hold.

... implies $\ell \mid$...

+... implies $\ell^v \mid$...

... $\{25\}(t), \{10\}(t)$...

+... $\{25\}(t), \{25\}(t)$...

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```
-... m (t ...
+... m (t) ...

-... / e^2 \bbZ)^2$! ...
+... / e^3 \bbZ)^2$! ...
```

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```
-... \tmd(A(a, b), B(a, ...
+... \tmd(\A m(a, b), \B m(a, ...
```

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```
-... of \ref{Lemma: ...
+... of \Cref{Lemma: ...

-... \tg {10} ...
+... \tg {m} ...

-... \tf {10} ...
+... \tf {m} ...

-\tf {25} (t_1) \equiv \tg {25} (t_1) \equiv 0 \pmod {5^9} \ \ \text{but} \ \ \tg {25} (t_1) \not\equiv 0 \pmod {5^9}.
+\tf {25} (t_1) \equiv \tg {25} (t_1) \equiv 0 \pmod {5^9} \ \ \text{but} \ \ \tf {25} (t_1) \not\equiv 0 \pmod {5^{10}}.
```

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```
-Finally, part (f). From (d)--(e) we conclude that for  $m = 5, 10$ , we have
+Finally, part (f). From (d)--(e), for  $m = 5, 10$  the local factor at  $5^5$  is at most  $3126$ , the local factor at  $2^5$  is at most  $2^5$ ,
, and every other local factor is at most  $2^5$ . Thus

-\cT {m}(e) \leq 3126 \cdot \prod_{\substack{\ell \mid e \\ \ell \neq 5}} \left(1 + \frac{1}{\ell}\right) \leq 1563 \cdot 2^{\omega(e)},
-\end{equation}
-and for  $m = 25$  we have
+\cT {m}(e) \leq 2 \cdot 3126 \cdot 2^{\omega(e)}.
+\end{equation}
+Similarly, for  $m = 25$ , the local factors at  $2^5$  and  $5^5$  are at most  $8$  and  $1953126$ , respectively, so

-\cT {25}(e) \leq 8 \cdot 1953126 \cdot \prod_{\substack{\ell \mid e \\ \ell \neq 2, 5}} \left(1 + \frac{1}{\ell}\right) \leq
7812504 \cdot 2^{\omega(e)},
+\cT {25}(e) \leq 8 \cdot 1953126 \cdot 2^{\omega(e)}.

-%If  $3^5$  or  $7^5$  divides  $e$ , we at worst introduce a factor of  $(27/2)(823544/2)=5558922$ , so in the general case, we see
-%\begin{equation}
-%T(e) \leq \frac{27}{2} \cdot 823544 \cdot 2^{\omega(e)} = 5558922 \cdot 2^{\omega(e)} = O(2^{\omega(e)})
-%\end{equation}
-so  $\cT m(e) = O(2^{\omega(e)})$  as claimed.
+Hence  $\cT m(e) = O(2^{\omega(e)})$  as claimed.
```

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```
- \repr 5 = \repr {10} : (a, b) \mapsto \begin{pmatrix}
- a & 2b \\
+ \repr {25} : (a, b) \mapsto \begin{pmatrix}
+ a & b \end{pmatrix}
```

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```
-&225000000 a^{16} - 768750000 a^{14} + 1004103125 a^{12} \ \ \&- 601050000 a^{10} + 139912500 \dots \&- 2 \cdot 3 \cdot 89 \cdot 8429 \cdot
\cdot b^2 + 5^2 \cdot 17^2
+ &253125 a^{12} - 478125 a^{10} + 480625 a^8 - 455000 a^6 \ \ \
+ &+ 252200 a^4 - 49360 a^2 + 16 \ \ \
+ = &3^4 \cdot 5^5 \cdot a^{12} - 3^2 \cdot 5^5 \cdot 17 \cdot a^{10} + 5^4 \cdot 769 \cdot a^8 \ \ \
+ &- 2^3 \cdot 5^4 \cdot 7 \cdot 13 \cdot a^6 + 2^3 \cdot 5^2 \cdot 13 \cdot 97 \cdot a^4 - 2^4 \cdot 5
\cdot 617 \cdot a^2 + 2^4, \ \ \text{and} \ \ \
+ &253125 b^{12} - 1040625 b^{10} + 1886875 b^8 - 1748750 b^6 \ \ \
+ &+ 786575 b^4 - 140665 b^2 + 3481 \ \ \
+ = &3^4 \cdot 5^5 \cdot b^{12} - 3^2 \cdot 5^5 \cdot 37 \cdot b^{10} + 5^4 \cdot 3019 \cdot b^8 \ \ \
+ &- 2 \cdot 5^4 \cdot 1399 \cdot b^6 + 5^2 \cdot 73 \cdot 431 \cdot b^4 - 5 \cdot 7 \cdot 4019 \cdot b^2 +
59^2
```

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-... 10, 25}\$, the arguments ...
+... 10, 25}\$, consider the arguments ...

- $\text{rawheight}(\text{A}_{\{m\}}(a, b), \text{B}_{\{m\}}(a, b)) / \text{C}_{\{10\}}(a, b)^6$
+ $\text{rawheight}(\text{A}_{\{m\}}(a, b), \text{B}_{\{m\}}(a, b)) / \text{C}_{\{m\}}(a, b)^{\deg B_{\{m\}}}$.

- have $27 |\text{B}_{\{m\}}(a, b)|^2 > 4 |\text{A}_{\{m\}}(a, b)|^3$. For the reader's information,
+ At these arguments, we have $27 |\text{B}_{\{5\}}(a, b)|^2 > 4 |\text{A}_{\{5\}}(a, b)|^3$ when $m = 5$, $4 |\text{A}_{\{10\}}(a, b)|^3 > 27 |\text{B}_{\{10\}}(a, b)|^2$ when $m = 10$, and $4 |\text{A}_{\{25\}}(a, b)|^3 = 27 |\text{B}_{\{25\}}(a, b)|^2$ when $m = 25$. For the reader's information,

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-... be a m -groomed ...
+... be an m -groomed ...

-where $5^6 / 2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}} = 7\,160.050\ldots$, and
+where $5^6 / (2^{\frac{1}{3}} \cdot 3^{\frac{1}{2}}) = 7\,160.050\ldots$, and

-... $\frac{2^{\frac{211}{90}} \cdot 5^6}{3^{\frac{1}{12}}}$...
+... $\frac{2^{\frac{7}{3}} \cdot 5^6}{3^{\frac{1}{10}}}$...

-where $2^{\frac{211}{90}} \cdot 5^6 / 3^{\frac{1}{12}} = 72\,411.579\ldots$.
+where $2^{\frac{7}{3}} \cdot 5^6 / 3^{\frac{1}{10}} = 70\,552.307\ldots$.

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-... $25(c)$,
+... $25(b)$,

-... admitting a m -isogeny ...
+... admitting an m -isogeny ...

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-... $X^{\frac{1}{18}}$.
+... $X^{\frac{1}{30}}$.

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-... $(X; e) = \sum_{\substack{d \mid 11 \\ X^{\frac{1}{2} \deg B_{\{m\}}} \mid \gcd(d, e)}} \dots$
+... $(X; e) = \sum_{\substack{d \mid 11 \\ X^{\frac{1}{2} \deg B_{\{m\}}} \mid \gcd(d, e)}} \dots$

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-... $X^{\frac{1}{2 \deg B_{\{m\}}}} \{d \dots$
+... $X^{\frac{1}{2 \deg B_{\{m\}}}} \{d \dots$

-... $X, d, e \geq 1$.
+... $X, d, e \geq 1$,

-... $0 \left(\frac{2^{\omega(e)}}{X^{\frac{1}{2 \deg B_{\{m\}}}} \right) \dots$
+... $0 \left(\frac{2^{\omega(e)}}{X^{\frac{1}{2 \deg B_{\{m\}}}} \right) \dots$

- for $X \geq 2$, $d, e \geq 1$.
+ for $X \geq 2$ and $e \geq 1$.

-... partial proof. ...
+... partial proofs. ...

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-... $\{e^2\}, (a, b) \dots$
+... $\{e^2\}, (a:b) \dots$

-... $\left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$
+... $\left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$

-... $0 \left(\frac{X^{\frac{1}{2 \deg B_{\{m\}}}}}{e^2} \right) \dots$

```
+... 0\parent{\frac{X^{1/(2 \deg B m)}}{e^2}} ...

-... X^{1/2 \deg B m},
+... X^{1/(2 \deg B m)},

-... X^{1/2 \deg B m}\{d\}.
+... X^{1/(2 \deg B m)}\{d\}.
```

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```
-... m(x; e) = & \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... m(X; e) = & \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... \frac{1}{\ell} \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... \frac{1}{\ell} \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... \frac{X^{1/2 \deg B m}}{\ell} \sum_{\substack{d \mid 11 X^{1/2 \deg B m} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...
+... \frac{X^{1/(2 \deg B m)}}{\ell} \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \\ \gcd(d, e) = 1}} X^{1/(2 \deg B m)} ...

-... 0(X^{-1/2 \deg B m})
+... 0(X^{-1/(2 \deg B m)})

-... X^{1/2 \deg B m}
+... X^{1/(2 \deg B m)}

-... m(x; e) ...
+... m(X; e) ...

-... 0(2^{\omega(e)} X^{1/2 \deg B m}) ...
+... 0(2^{\omega(e)} X^{1/(2 \deg B m)}) ...
```

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```
- Let $m \in 10, 25$. Suppose $y \mid 11 X^{1/2 \deg B m}$. Then
+ Let $m \in \{10, 25\}$. Suppose $2 \leq y \leq X$. Then

-... m \log y \{y^{1-6/\deg B m}, X^{1/2 \deg B m} y^{3/\deg B m} \log X \log^2 y\}
+... m \log y \{y^{1-6/\deg B m}, X^{1/(2 \deg B m)} y^{3/\deg B m} \log X \log^2 y\}

-... e^{3/\deg B m} X^{1/2 \deg B m} \log X\{n\}.
+... e^{3/\deg B m} X^{1/(2 \deg B m)} \log X\{n\}.

-... e^{3/\deg B m} X^{1/2 \deg B m} \log X\{n\}.
+... e^{3/\deg B m} X^{1/(2 \deg B m)} \log X\{n\}.
```

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```
-... \frac{1}{\ell} = 0\parent{\frac{2^{\omega(n)}}{n^{2-6/\deg B m}}} = 0\parent{\sum_{n > ...} \frac{1}{\ell}} = 0\parent{\sum_{n > ...} \frac{1}{\ell}}

-... &= \frac{R^7 X^{1/6}}{m^{1/\deg B m} \zeta(2)} \parent{Q\{m\}} ...
+... &= \frac{R^m X^{1/\deg B m}}{m^{1/\deg B m} \zeta(2)} \parent{Q\{m\}} ...

-... X^{1/6} + 0\parent{\frac{X^{1/6} \log y}{y^{1-\deg B m/6}}}
+... X^{1/\deg B m} + 0\parent{\frac{X^{1/\deg B m} \log y}{y^{1-6/\deg B m}}}

-As the summands of \eqref{equation: sum for Q for m = 7} constitute a nonnegative multiplicative arithmetic function, we can factor
$Q\{m\}$ ...
+As the summands of \eqref{equation: sum for Q for m = 10 and 25} constitute a nonnegative multiplicative arithmetic function, we can
factor $Q\{m\}$ ...
```

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```
-... &\colonequals \sum_{a \geq 0} \frac{\varphi_6(\deg B a)}{a^{1/\deg B m}} ...
+... &\colonequals 1 + \sum_{a \geq 1} \frac{\varphi_6(\deg B a)}{a^{1/\deg B m}} ...

-... \left(2+2^{2/3}\right), & \text{if } m = 25 \text{ and } p=5; \}
```

```

+... \left(2+2^{2/3}\right), & \text{if } \$m = 25\$ and \$p=2\$; \\\
-... {10}(2^2)}{\parent{1 ...
+... {25}(2^2)}{\parent{1 ...

-... \ll X^{1/2 \degB m}\$, ...
+... \leq X\$, ...

-      &\sum_{n \leq y} \sum_{e \mid n} \mu \parent{n/e} 0\parent{\frac{2^{\omega(n)}}{e^{3/\degB m}} X^{1/2 \degB m} \log X}\{n
  }\} \\\
-      =& 0\parent{X^{1/2 \degB m} \log X \sum_{e \leq y} \frac{2^{\omega(e)}}{e^{1-3/\degB m}} \sum_{f \leq y/e} \frac{2^{\omega(f)}}{f}}.
+      &\sum_{n \leq y} \sum_{e \mid n} \mu \parent{n/e} 0\parent{\frac{2^{\omega(n)}}{e^{3/\degB m}} X^{1/(2 \degB m)} \log X
  }\{n\}} \\\
+      =& 0\parent{X^{1/(2 \degB m)} \log X \sum_{e \leq y} \frac{2^{\omega(e)}}{e^{1-3/\degB m}} \sum_{f \leq y/e} \frac{2^{\omega(f)}}{f}}.

```

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```

-&0\parent{X^{1/2 \degB m} ...
+&0\parent{X^{1/(2 \degB m)} ...

-... \frac{2^{\omega(f)}}{f}} \\\ =& 0\parent{X^{1/2 \degB m} \log X ...
+... \frac{2^{\omega(f)}}{f}} \\\ =& 0\parent{X^{1/(2 \degB m)} \log X ...

-... =& 0\parent{X^{1/2 \degB m} y^{3/\degB ...
+... =& 0\parent{X^{1/(2 \degB m)} y^{3/\degB ...

-... p^{6/\degB m}}}, \\\ %\sum_{n \geq ...
+... p^{6/\degB m}}}, %\sum_{n \geq ...

```

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```

-      Now by \Cref{Proposition: bounding the summands of twN calE X}, we see
+      Set  $\alpha=6/\degB m$ . By \Cref{Proposition: bounding the summands of twN calE X},

-      \twistNadgy m(X) = 0\parent{\sum_{n > y} \frac{2^{\omega(n)}}{X^{1/\degB m}}\{n^{2 - 6/\degB m}\}}.
+\begin{aligned}
+      &\abs{\twistNadgy m(X)} \\
+      &\ll \sum_{n > y} \sum_{e \mid n} \abs{\mu(n/e)} \\
+      &\frac{2^{\omega(n)}}{e^6 X^{1/\degB m}}\{n^2\} \\\
+      &= X^{1/\degB m} \sum_{n > y} \\
+      &\frac{2^{\omega(n)}}{n^{2-\alpha}} \\
+      &\prod_{p \mid n} \parent{1+p^{-\alpha}}.
+\end{aligned}

-      Combining \Cref{Corollary: sum of 2^omega(n)} and \Cref{Corollary: tail of sum of f/n^sigma}, we conclude
+      Since  $m=10, 25$  gives  $0 < \alpha < 1$ , \Cref{Lemma: weighted 2omega divisor factor} with  $\beta=2-\alpha > 1$  yields

-      \twistNad m(X) = \twconst m X^{1/\degB m} + 0\parent{X^{1/2(\degB m - 3)} \log^{(\degB m + 3) / (\degB m - 3)} X}
+      \twistNad m(X) = \twconst m X^{1/\degB m} + 0\parent{X^{1/(2(\degB m - 3))} \log^{3(\degB m - 5)/(\degB m - 3)} X}

-      Let  $m = 10, 25$ , and let  $\$y\$$  be a positive quantity with  $\$y \ll X^{1/2 \degB m}\$$ ; in particular,  $\$\log y \ll \log X\$$ . \Cref{
  Lemma: asymptotic for twN<=y(X) for m = 10 and 25} and \Cref{Lemma: bound on twN>y(X) for m = 10 and 25} together tell us
+      Let  $m = 10, 25$ . \Cref{Lemma: asymptotic for twN<=y(X) for m = 10 and 25} and \Cref{Lemma: bound on twN>y(X) for m = 10 and
  25} together tell us

```

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```

-\twistNad 7(X) = \twconst m ...
+\twistNad m(X) = \twconst m ...

-... m\log y\{y^{1 - 6/\degB m}\}, X^{1/2 \degB m} y^{3/\degB m} \log X \log^2 y \}.
+... m\log y\{y^{1 - 6/\degB m}\}, X^{1/(2 \degB m)} y^{3/\degB m} \log X \log^2 y \}.

-      We let  $y = X^{1/(2(\degB m - 3))} \log^2 \degB m / (\degB m - 3) X$ , so
+      We let

-      \frac{X^{1/\degB m} \log y\{y^{1 - 6/\degB m}\} \asymp X^{1/2 \degB m} y^{3/\degB m} \log X \log^2 y \asymp X^{1/(2(\degB m - 3))}
  \log^{(\degB m + 3) / (\degB m - 3)} X,
+      y = X^{1/(2(\degB m - 3))} \log^{-2 \degB m} / (\degB m - 3) X.

-      and we conclude
+      For sufficiently large  $\$X\$$ , this choice satisfies  $\$2 \leq y \leq X\$$ . Moreover,  $\$\log y \asymp \log X\$$ , and a direct
  calculation gives

-      \twistNad m(X) = \twconst m X^{1/\degB m} + 0\parent{X^{1/2(\degB m - 3)} \log^{(\degB m + 3) / (\degB m - 3)} X}

```

```

+ \frac{X^{1/\deg B m} \log y}{y^{1-6/\deg B m}} \asymp X^{1/(2 \deg B m)} y^{3/\deg B m} \log X \log^2 y \asymp X^{1/(2(\deg B m - 3))} \log^3(\deg B m - 5)/(\deg B m - 3) X.

- as desired.
+ The finitely many smaller values of  $\beta$  may be absorbed into the implicit constant, proving the claim.

```

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```

-\subsection*{$L$-series}\label{Subection: L-series for m = 10 and 25}\label{Subsection: L-series for m = 10 and 25}
+\subsection*{$L$-series}\label{Subsection: L-series for m = 10 and 25}

- Let  $m = 10, 25$ . The following statements hold.
+ Let  $m = 10, 25$ , and put
+ \begin{equation}
+ \beta_m \colonequals \frac{1}{2(\deg B m - 3)}.
+ \end{equation}
+ The following statements hold.

-...  $1/2(\deg B m - 3)$ . ...
+...  $\beta_m$ . ...

```

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```

- \item We have
-\begin{equation}
- \mu_{\text{twistLad } m}(\sigma) < 13/84
- \end{equation}
- for  $\sigma > 1/(2 \deg B m - 3)$ .
+ \item For  $\sigma > \beta_m$ , we have
+ \begin{equation}
+ \mu_{\text{twistLad } m}(\sigma) \leq \max\left\{0, \frac{1}{\deg B m - \sigma} - \frac{1}{\deg B m - \beta_m}\right\}.
+ \end{equation}

- The proof is structurally identical to the one given for \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7}.
+ The proof is structurally identical to the one given for \Cref{Corollary: twL(s) has a meromorphic continuation for m = 7},
with  $1/6$  and  $1/12$  replaced by  $1/\deg B m$  and  $\beta_m$ .

-... \ref{Section: ...}
+... \Cref{Section: ...}

```

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```

-... For (a), since  $\zeta(s)$  is nonvanishing when  $\sigma > 1$ , the ratio  $\zeta(6s)/\zeta(12s)$  is meromorphic function for  $\sigma > 1/12$ . But \Cref{Corollary: twL(s) has a meromorphic continuation ...}
+... For (a), since  $\zeta(s)$  is nonvanishing when  $\sigma > 1$ , the ratio  $\zeta(6s)/\zeta(12s)$  is a meromorphic function for  $\sigma > 1/12$ . But \Cref{Corollary: twL(s) has a meromorphic continuation ...}

```

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```

-we indeed have  $\mathcal{L}_{\text{Qad } 7}(n) \in \mathbb{Z}_{\geq 0}$ .
+we indeed have  $\mathcal{L}_{\text{Qad } m}(n) \in \mathbb{Z}_{\geq 0}$ .

- and by \Cref{Theorem: product of Dirichlet series converges} this series converges absolutely for  $\sigma > 1/6$ , so the
abscissa of absolute convergence for  $\mathcal{L}_{\text{Qad } m}(s)$  is at most  $1/6$ . But for  $\sigma < 1/6$ , we have
+ and by \Cref{Theorem: product of Dirichlet series converges} this series converges absolutely for  $\sigma > 1/6$ , so the
abscissa of absolute convergence for  $\mathcal{L}_{\text{Qad } m}(s)$  is at most  $1/6$ . For the reverse inequality, choose any twist-minimal curve
in the family and let  $n_0$  be its twist height. By \Cref{Theorem: relationship between twistL(s) and L(s)}(a), the
coefficients of  $\mathcal{L}_{\text{Qad } m}(s)$  satisfy
+ \begin{equation}
+ \mathcal{L}_{\text{Qad } m}(n_0 c^6) \geq 2 \text{abs}(\mu(c))
+ \end{equation}
+ for every positive integer  $c$ . Hence the Dirichlet series for  $\mathcal{L}_{\text{Qad } m}(s)$  contains the divergent positive subseries

-  $\mathcal{L}_{\text{Qad } m}(\sigma) > \frac{2 \zeta(6s)}{\zeta(12s)}$ 
+  $2n_0^{-\sigma} \sum_{c \geq 1} \frac{1}{c^{\sigma}} \text{abs}(\mu(c)) c^{6\sigma}$ 

- by termwise comparison of coefficients, so the Dirichlet series for  $\mathcal{L}_{\text{Qad } m}(s)$  diverges when  $\sigma < 1/6$ , and (i) holds with  $\sigma_a = 1/6$ .
+ whenever  $\sigma \leq 1/6$ . Thus its abscissa of absolute convergence is at least  $1/6$ , and (i) holds with  $\sigma_a = 1/6$ .

-...  $\zeta(6s)/\zeta(12s)$  has a meromorphic contintuation to  $\sigma > 1/12$ , and so ...
+...  $\zeta(6s)/\zeta(12s)$  has a meromorphic continuation to  $\sigma > 1/12$ , and so ...

```

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... is absolutely convergent for $s > 1/12$, \Cref{Theorem: absolutely ...}
 +... is absolutely convergent for $\sigma > 1/12$, \Cref{Theorem: absolutely ...}

... $m(n)_{n \geq 1}$ is admissible with parameters $\left(1/6, 1/12, 13/42\right)$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to ...
 +... $m(n)_{n \geq 1}$ is admissible with parameters $\left(1/6, 1/12, 13/84\right)$. We now apply \Cref{Theorem: Landau's Tauberian theorem} to ...

... most $O(X^{1/2} \deg B m \dots)$
 +... most $O(X^{1/(2 \deg B m)} \dots)$

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-The algorithm described in \cref{Section: Computations for $m = 7$ } can be adapted to ... $b_m) = m$ and $C(a_e, b_e) = e_0$, but also products of their conjugates.
 +The algorithm described in \cref{Section: Computations for $m = 7$ } can be adapted to enumerate elliptic curves admitting a cyclic m -isogeny for $m = 10, 25$. Doing so requires paying special attention to the primes 2 and 5 , rather than 3 and 7 , and requires writing $C_m(a, b) = e_0^2 n_0$, with n_0 squarefree. In step 3, if e_0^2 lies in the lookup range, we multiply representatives of norm e_0^2 by representatives of norm n_0 ; otherwise we square representatives of norm e_0 first. When $m = 10$, the lookup table in step 2 is restricted to pairs (a, b) with $a \geq b$; in step 3 we also use their conjugates. The bounded components at 2 and 5 are handled separately.

-For $m = 10$, running our algorithm out to $X = 10^{96}$ in Python took us approximately 17 CPU hours 61613.3902490139 seconds
 -on a single core, producing $106\,785\,277$ elliptic curves admitting a cyclic 10 -isogeny. To check the accuracy of our code, we confirmed that the j -invariants of these curves are distinct. For $X = 10^{96}$, we have
 +For $m = 10$, enumerating out to $X = 10^{96}$ produces $106\,785\,276$ elliptic curves admitting a cyclic 10 -isogeny. For $X = 10^{96}$, we have

-For $m = 25$, running our algorithm out to $X = 10^{138}$ took us approximately 10 CPU hours 36199.61623263359 seconds
 -on a single core, producing $34\,908\,299$ elliptic curves admitting a cyclic 25 -isogeny. To check the accuracy of our code, we confirmed that the j -invariants of these curves are distinct. For $X = 10^{138}$, we have
 +For $m = 25$, enumerating out to $X = 10^{138}$ produces $35\,007\,063$ elliptic curves admitting a cyclic 25 -isogeny. For $X = 10^{138}$, we have

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... $0.997\,115\dots$, \label{Equation: ...}
 +... $0.999\,937\dots$, \label{Equation: ...}

... compute $\text{twconst}\{10\}$ below. It ...
 +... compute $\text{twconst}\{25\}$ below. It ...

-We list the first few twist minimal elliptic curves admitting a cyclic 10 -isogeny ...
 +We list some twist minimal elliptic curves admitting a cyclic 10 -isogeny ...

-\{ E \in \text{twistE} with a cyclic 10 -isogeny ...
 +\{Some E \in \text{twistE} with a cyclic 10 -isogeny ...

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... 25 as a product over $p \leq Y$ gives us a lower bound
 +... 25 as a product over $p \leq Y$ gives the lower bound

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... appearing in \Cref{Theorem: asymptotic for $NQ(X)$ for $m = 10$...
 +... appearing in \Cref{Theorem: asymptotic for $twN(X)$ for $m = 10$...

... $\frac{13}{6} \cdot 63 \cdot 8 \prod_{\substack{p \dots \\ p \leq Y, \dots}}$
 +... $\prod_{\substack{p \dots \\ p \leq Y, \dots}}$

... $\frac{2p}{p^6 \deg B m - 1} \cdot \frac{(p+1)}{(p+1)p^2 - p^6 \deg B m}$
 +... $\frac{2p}{p^6 \deg B m - 1} \cdot \frac{(p+1)}{(p+1)p^2 - p^6 \deg B m}$

-for Q_m . The values $Q_m(2)$ and $Q_m(5)$ are recorded ... Recall \eqref{Equation: prime counting function}. Using Abel summation and Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer}, we obtain
 +Put $\alpha_m = 6 \deg B m$ and $s_m = 2 - \alpha_m$. For $p > Y$, the nontrivial local increment is less than $2p^{-s_m} / (1 - Y^{-s_m})$. Thus, by Abel summation and Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer}, for $Y \geq 8 \cdot 10^9$,

-\begin{aligned} \sum_{\substack{p > Y \\ p \equiv 1 \pmod{4}}} \frac{1}{p^2 + 1} &= -\frac{\pi(Y; 4, 1)}{Y^2 + \dots} + \log Y + \frac{1}{Y} + \frac{1}{2 \log^2 Y} + \frac{1}{\pi^2} - \tan^{-1}(Y) \\ Q_m &< Q_{\lfloor m \rfloor} \exp \left(\frac{Y^{1-s_m}}{(s_m-1) \log Y} \right) \end{aligned}

+\\parent{1+\\frac{5s_m}{2\\log Y}}.

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-In particular, letting $Y = 10^{11}$, we compute
+In particular, taking $Y = 10^{11}$ gives

-3.636\\,493\\,079\\,001\\,437\\,6 < \\Q {10} < 3.636\\,493\\,079\\,020\\,102, %52509.778564453125 seconds %15 CPU hours, $Y = 10^{11}$
+3.636\\,493\\,079\\,001\\,437\\,6 < \\Q {10} < 3.636\\,494\\,121\\,479\\,969

-4.244\\,853\\,881\\,138\\,272\\,6 < \\Q {25} < 4.244\\,853\\,881\\,160\\,06; %53808.07576751709 seconds %2 CPU hours, $Y = 10^{11}$
+4.244\\,853\\,881\\,138\\,272\\,6 < \\Q {25} < 4.244\\,853\\,894\\,726\\,220.

-these estimates require approximately \$15\$ CPU hours apiece.

... R_{10} , ...
+... $\text{calR}_{10}(1)$, ...

-1.141\\,059\\,04 \\cdot \\sqrt{\\frac{r_{10}(s_{10}) - r_{10}}{s_{10}^3}} < 1.6 \\cdot 10^{-8}.
+1.141\\,059\\,04 \\cdot \\sqrt{\\frac{r_{10}(s_{10}) - r_{10}}{s_{10}^3}}
+=1.516\\,12\\ldots \\cdot 10^{-6}.

... R_{25} , ...
+... $\text{calR}_{25}(1)$, ...

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-We therefore have
-\\begin{equation} \\begin{aligned} & \\twconst {10} \\approx 1.068\\,204 \\cdot \\text{with error bounded by} \\cdot 3.4 \\cdot 10^{-6}, \\ \\ & \\twconst {25} \\approx 0.754\\,252\\,0 \\cdot \\text{with error bounded by} \\cdot 9.6 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+Using the finite Euler products as point estimates, we therefore have
+\\begin{equation} \\begin{aligned} & \\twconst {10} \\approx 1.068\\,203\\,87 \\cdot \\text{with Monte Carlo standard error } 3.36 \\cdot 10^{-6}, \\ \\ & \\twconst {25} \\approx 0.754\\,251\\,990 \\cdot \\text{with Monte Carlo standard error } 9.58 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+The corresponding errors from truncating the Euler products are below $3.07 \\cdot 10^{-7}$ and $2.42 \\cdot 10^{-9}$, respectively.

-In this subsection, we estimate $\\constad {10} = 2 \\twistNad {10} (1/6) / \\zeta(2)$ and $\\constad {25} = 2 \\twistNad {10} (1/6) / \\zeta(2)$, the constants which appear in \\Cref{Theorem: asymptotic for NQ(X) for m = 10 and 25}, by computing the partial sums of $\\twistLad {10} (1/6)$ and $\\twistLad {25} (1/6)$:
+In this subsection, we estimate $\\constad {10} = 2 \\twistLad {10} (1/6) / \\zeta(2)$ and $\\constad {25} = 2 \\twistLad {25} (1/6) / \\zeta(2)$, the constants which appear in \\Cref{Theorem: asymptotic for NQ(X) for m = 10 and 25}, using the partial sums

... 0.869\\,838\\,621\\,652\\,207\\,3, ...
+... 0.411\\,595\\,409\\,503\\,702\\,919, ...

... 0.206\\,338\\,924\\,690\\,954\\,36.
+... 0.206\\,338\\,924\\,692\\,701\\,754.

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-If these bounds continue to hold for larger Y , then \\begin{equation} \\begin{aligned} \\sum_{n > ...} \\constad {25} & \\approx 0.25087816 \\cdot \\text{with error bounded by} \\cdot 1.2 \\cdot 10^{-7}. \\ \\ \\end{aligned} \\end{equation}

+If these bounds continue to hold for larger Y , partial summation gives
+\\begin{equation} \\begin{aligned} & \\sum_{n > Y} \\frac{\\twisthad m(n)}{n^{1/6}} \\ \\ & + -Y^{-1/6} \\twistNad m(Y) + \\frac{1}{6} \\int_Y^{\\infty} \\twistNad m(x) x^{-7/6} dx. \\ \\ \\end{aligned} \\end{equation}

+Using $\\twistNad {10} (10^{96}) = 106\\,785\\,276$ and $\\twistNad {25} (10^{138}) = 35\\,007\\,063$, the two tails are respectively less than $1.12266724 \\cdot 10^{-8}$ and $2.83347 \\cdot 10^{-16}$. Hence
+\\begin{equation} \\begin{aligned} & \\twistLad {10} (1/6) \\approx 0.411595415117039 \\cdot \\text{with conditional error } < 5.62 \\cdot 10^{-9}, \\ \\ & \\twistLad {25} (1/6) \\approx 0.20633892469270190 \\cdot \\text{with conditional error } < 1.42 \\cdot 10^{-16}, \\ \\ \\end{aligned} \\end{equation}

+and therefore
+\\begin{equation} \\begin{aligned} & \\constad {10} \\approx 0.500440015697013 \\cdot \\text{with conditional error } < 6.83 \\cdot 10^{-9}, \\ \\ & \\constad {25} \\approx 0.25087804897622103 \\cdot \\text{with conditional error } < 1.73 \\cdot 10^{-16}. \\ \\ \\end{aligned} \\end{equation}

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... 13\$, with an improved error term. ...
+... 13\$.

-... theoretical estimate
+... theoretical estimate.

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-
$$\mathbb{Z} = 2 (a^6 + 512 a^5 b - 13073 a^4 + 34860 a^3 - 157099 a^2 + 211330 a - 655108).$$

+
$$\mathbb{Z} = 2 (a^6 + 512 a^5 b - 13073 a^4 b^2 + 34860 a^3 b^3 - 157099 a^2 b^4 + 211330 a b^5 - 655108 b^6).$$

-... b) $\{C_{13}\}$...
+... b) $\{C_{II}^{13}\}$...

-... $\mathbb{Z}$ -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{\geq 0}$. Then the following statements hold.
+... $\mathbb{Z}$ -groomed, let ℓ be prime, and let $v \in \mathbb{Z}_{>0}$. Then the following statements hold.

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-... and 25} however.
+... and 25}, however.

-... the minimality defect $\text{\eqref{Equation: ...}}$
+... the twist minimality defect $\text{\eqref{Equation: twist ...}}$

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-... $\{13\}(t,1)=\text{\Res}(\text{\textit{tA}} \dots$
+... $\{13\}(t,1)=\text{\Res}(\text{\textit{tf}} \dots$

-This proves (a)
+This proves (a).

-... $\{13\}(b,0) \dots$
+... $\{13\}(a,0) \dots$

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-... 3 \mid \text{\textit{hIII}}^{13} (a, \dots

+... 3 \mid \text{\textit{CIII}}^{13} (a, \dots

-... 13 lesser} is ...
+... 13} are ...

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-... $e_1 = 3^{v_1} 13^{w_1} \dots$
+... $e_1 = 2^{u_1} 3^{v_1} 13^{w_1} \dots$

-... $13^{w_2} e_2 \text{\textit{prm}}$, where $\gcd(e_1 \text{\textit{prm}}, \dots$
+... $3^{v_2} 13^{w_2} e_2 \text{\textit{prm}}$, where $\gcd(e_1 \text{\textit{prm}}, 2 \cdot \dots$

-... $\gcd(e_2 \text{\textit{prm}}, 2 \cdot 13) \dots$
+... $\gcd(e_2 \text{\textit{prm}}, 2 \cdot 3 \cdot 13) \dots$

-... $(a, b) \text{\textit{psmod}} \{(e_2 \text{\textit{prm}})^2\}$,
+... $(a, b) \text{\textit{equiv}} 0 \text{\textit{psmod}} \{(e_2 \text{\textit{prm}})^2\}$,

-... $w_2 > 0$ then $A^{13} (a, b) \text{\textit{equiv}} 0 \text{\textit{psmod}} \{13^{2w_1}\}$ and $B^{13} (a, b) \text{\textit{equiv}} 0 \text{\textit{psmod}} \{13^{3w_1}\}$, ...
+... $w_2 > 0$ then $A^{13} (a, b) \text{\textit{equiv}} 0 \text{\textit{psmod}} \{13^{2w_2}\}$ and $B^{13} (a, b) \text{\textit{equiv}} 0 \text{\textit{psmod}} \{13^{3w_2}\}$, ...

-... $\text{\textit{psmod}} \{13^{3w_1}\}$, but $C_{II} \dots$
+... $\text{\textit{psmod}} \{13^{3w_2}\}$, but $C_{II} \dots$

-... $\not\equiv 0 \text{\textit{psmod}} \{13\}$.
+... $\equiv 0 \text{\textit{psmod}} \{13^2\}$.

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-\item For all prime $\ell \neq 2$, ...
+\item For every prime $\ell \neq 2$, ...

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```

-... = 0(2^{\omega(e_1} ...
+... = 0(2^{\omega(e_1} ...

-      For parts (a) and (b), by the CRT (Sun Zi theorem), it suffices to consider  $e = \ell^v$  a power of a prime. For  $\ell \neq$ 
2, 3, 13, both claims follow from \Cref{Lemma: 2 and 3 and 13 are the ungroomed primes for  $m = 13$ }(a)–(b). But a finite
computation verifies our claim in these cases as well (see the proof of (e) below).
+      For parts (a) and (b), by the CRT (Sun Zi theorem), it suffices to consider  $e = \ell^v$  a power of a prime. For  $\ell \neq$ 
2, 3, 13, both claims follow from \Cref{Lemma: 2 and 3 and 13 are the ungroomed primes for  $m = 13$ }(a)–(b). For the
exceptional primes, the stated conditions follow directly from \Cref{Definition T(e) for  $m = 13$ } and \eqref{Equation: twist
defect powers}, apart from the two assertions that no pairs are permitted when  $2 \mid e_1$  or  $3 \mid e_2$ . The first follows
by checking the defining conditions modulo 8, and the second by checking them modulo 27. Part (b) then follows from the
divisibility conditions in part (a).

-We observe  $A(1, 0) = -3$  and  $B(1, 0) = 2$  are ...
+We observe  $A_{13}(1, 0) = -3$  and  $B_{13}(1, 0) = 2$  are ...

-... as  $A(1, 0) = -3$  and  $B(1, \dots$ 
+... as  $A_{13}(1, 0) = -3$  and  $B_{13}(1, \dots$ 

```

Submitted Thesis pp. 208–209 (deliberate completion)

```

+      For  $v \geq 1$ , the 13-adic values are
+      \begin{equation}\label{Equation: cT13 at 13}
+      \begin{cases}
+      1+12 \cdot 13^{3v-2}, & 1 \leq v \leq 3, \\
+      1+13^8, & v \geq 4,
+      \end{cases}
+      \quad
+      \begin{cases}
+      \begin{cases}
+      1+13^{2v-2}, & 1 \leq v \leq 3, \\
+      1+13^5, & v \geq 4.
+      \end{cases}
+      \end{cases}
+      \end{equation}
+
+It remains to treat  $\ell=13$ . Modulo 13 we have
+\begin{equation}
+\begin{aligned}
+&f_{13}(t) \equiv -3(t-2)(t-3)(t+3)^6, \\
+&g_{13}(t) \equiv 2(t-2)(t-3)^2(t+3)^9.
+ \end{aligned}
+\end{equation}
+Thus only the residue classes  $t \equiv 2, 3, 10 \pmod{13}$  can contribute. At  $t \equiv 2$ , the factor  $h_{II,13}$  has a simple root
while  $h_{III,13}$ ,  $t_{13}$ , and  $tg_{13}$  are units, so Hensel's lemma gives exactly one class in  $\mathcal{T}_{13}(13^v, 1)$  for
every  $v$ . Likewise, the simple root  $t \equiv 3$  of  $h_{III,13}$  gives exactly one class in  $\mathcal{T}_{13}(1, 13^v)$  for every  $v$ .
+
+For the common residue class, write  $t=10+13u$ . Then
+\begin{equation}
+\begin{aligned}
+&h_{II,13}(t) = 13H_{II}(u), \\
+&H_{II}(u) = 13u^2+21u+9, \\
+&h_{III,13}(t) = 13H_{III}(u), \\
+&H_{III}(u) = 13u^2+20u+8, \\
+&t_{13}(t) = 13^4F(u), \\
+&F(u) = -3(u^4-15u^3-31u^2-20u-4), \\
+&tg_{13}(t) = 13^6G(u), \\
+&G(u) = 2(u^6+44u^5+83u^4+20u^3-59u^2-50u-12).
+ \end{aligned}
+\end{equation}
+Modulo 13 these satisfy
+\begin{equation}
+H_{II}(u) \equiv 8(u-7), \quad \text{quad} \\
+H_{III}(u) \equiv 7(u-10), \quad \text{quad} \\
+F(u) \equiv -3(u-7)^4, \quad \text{quad} \\
+G(u) \equiv 2(u-10)^6.
+\end{equation}
+Consequently, for  $\mathcal{T}_{13}(13^v, 1)$  the defining convention forces  $H_{III}(u)$  to be a unit. If  $v \leq 3$ , the
displayed powers  $13^6$  and  $13^9$  already supply the required divisibility, giving  $(13-1)13^{3v-2}$  classes on this branch.
If  $v \geq 4$ , then  $G(u)$  is a unit and the condition on  $g_{13}$  is exactly
 $13^{3v-9} \mid H_{II}(u)$ . The latter polynomial has a simple root modulo 13, so there are exactly  $13^8$  such classes
modulo  $13^{3v}$ . Including the branch through  $t \equiv 2$  gives the first formula of \eqref{Equation: cT13 at 13}.
+
+For  $\mathcal{T}_{13}(1, 13^v)$ , the common branch must instead satisfy  $13 \mid H_{III}(u)$ , while  $H_{II}(u)$  and  $F(u)$ 
are units. A further substitution  $u=10+13w$  gives
+\begin{equation}
+13^{-1}G(10+13w) \equiv -5 \pmod{13},
+\end{equation}
+so  $v_{13}(G(u))=1$  throughout this branch. If  $r=v_{13}(H_{III}(u)) \geq 1$ , the two divisibility conditions are therefore

```

```

+\begin{equation}
+6+r\geq2v,\qquad 10+2r\geq3v.
+\end{equation}
+For  $v\leq3$  these impose only  $r\geq1$ , giving  $13^{2v-2}$  classes modulo  $13^{2v}$ ; for  $v\geq4$  they reduce to  $r\geq2v-6$ ,
giving  $13^5$  classes. Including the branch through  $t\equiv3$  gives the second formula of \eqref{Equation: cT13 at 13}.
-
-Notably, \Cref{Lemma: bound on T(e)} for  $m = 13$  does not furnish the values of  $cT_{13}(13^v, 1)$  and  $cT_{13}(1, 13^v)$  for  $v \geq 1$ . By Hensel's Lemma, these functions are constant for sufficiently large  $v$ , but our somewhat naive code runs into
memory issues before verifying these plateaued values. Unfortunately, this obstructs the computation of  $Q_{13}$  below;
however, it poses no issue for our theoretical results.

```

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```

-...  $h_{13}(t)$  ...
+... the corresponding polynomial ...

-...  $(2^{\ell_1}, 1) = 0$  for  $\ell_1$  ...
+...  $(2^v, 1) = 0$  for  $v$  ...

```

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```

-}{All nonzero  $cT_{13}(3^v, 1)$ }
+}{All nonzero  $cT_{13}(1, 2^v)$ }

-... 7} and \Cref{Theorem: asymptotic for NQ(X)} for  $m = 10$  and ...
+... 7} and \Cref{Theorem: Controlling size of twist minimality defect} for  $m = 5$  and  $10$  and ...

-... {II, 13}  $C_m(a, \dots)$ 
+... {II, 13}  $C_{II_{13}}(a, \dots)$ 

-... {III, 13}  $C_m(a, \dots)$ 
+... {III, 13}  $C_{III_{13}}(a, \dots)$ 

```

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```

-... If  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 
+... If  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_{\textup{II}}$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

-... evaluating  $C_{III}$  ...
+... evaluating  $C_{II}$  ...

```

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```

-... cases,  $27 |\mathbf{B}_m(a, b)|^2 > 4 |\mathbf{A}_m(a, b)|^3$ . For ...
+... cases,  $27 |\mathbf{B}_{13}(a, b)|^2 > 4 |\mathbf{A}_{13}(a, b)|^3$ . For ...

-... Write  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 
+... Write  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_{\textup{II}}$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

-...  $13(a)$  and , ...
+...  $13(a)$ , ...

-...  $u = v = w = \dots$ 
+...  $u, v, w \leq \dots$ 

```

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```

-...  $b + 1$ :
+...  $b^2 + 1$ .

-we maximize ...
+We maximize ...

-... minimize ...
+... minimize ...

-... analogues to \Cref{Theorem: Controlling size of twist minimality defect} for  $m = 13(b)$ .
+... analogous bounds.

-... if  $C_{II_{13}}(a, b) = e_{\textup{II}}^3 n_{\textup{II}}$ , with  $n_0$  cube-free, and  $C_{III_{13}}(a, b) = e_{\textup{III}}^2 \dots$ 

```

```
+... if $\CII {13} (a, b) = e_{\textup{II}}^3 n_{\textup{II}}$, with $n_{\textup{II}}$ cube-free, and $\CIII {13} (a, b) = e_{\textup{III}}^2 \dots$

-... 7^6} ...
+... 13^6} ...

-... \upperratio {13} ...
+... \upperratio {11, 13} ...
```

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```
-... X^{\frac{1}{18}}\{d^2\} ...
+... X^{\frac{1}{12}}\{d^2\} ...

-... 0\parent{\frac{2^{\omega(e)}}{X^{\frac{1}{24}}}\{d e^{\frac{3}{2}}\}}
+... 0\parent{\frac{2^{2 \omega(e)}}{X^{\frac{1}{24}}}\{d\}}
```

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```
-... $X \geq 2$ and $d, e \geq 1$.
+... $X \geq 2$ and $e \geq 1$.

-... \\\ $a/b$ ...
+... \\\ $(a:b)$ ...
```

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```
-      = & \R 7 \varphi(e_1^3 e_2^2) X^{\frac{1}{12}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\cT {13}(e_1, e_2)}{e_1^6 e_2^4} \\\
- &+ 0\parent{\varphi(e_1^3 e_2^2) X^{\frac{1}{24}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\cT {13}(e_1, e_2)}{e_1^3 e_2^2}} \\\
+      = & \R {13} X^{\frac{1}{12}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\varphi(e_1^3 e_2^2) \cT {13}(e_1, e_2)}{e_2^4} \\\
+ &+ 0\parent{X^{\frac{1}{24}} \sum_{\substack{e_1 e_2 = e \\ \gcd(e_1, e_2) = 1}} \frac{\varphi(e_1^3 e_2^2) \cT {13}(e_1, e_2)}{e_1^3 e_2^2}} \\\
```

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```
-... {13}(x; ...
+... {13}(X; ...

-... \frac{\R {13} X^{\frac{1}{12}}}{\prod_{\ell \mid e} \parent{1 - \frac{1}{\ell}}}} ...
+... \R {13} X^{\frac{1}{12}} \prod_{\ell \mid e} \parent{1 - \frac{1}{\ell}} ...

-\cM {13}(x;e)
+\cM {13}(X;e)
```

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```
-      \calR {13}(1) = \set{(a, b) \in \bbR^2 : \rawheight(\A m (a, b), \B m (a, b)) \leq 1, b \geq 0}.
+      \calR {13}(1) = \set{(a, b) \in \bbR^2 : \rawheight(\A {13} (a, b), \B {13} (a, b)) \leq 1, b \geq 0}.
```

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```
-... = \twconst {13} X^{\frac{1}{12}} + 0\parent{\max\parent{\frac{X^{\frac{1}{12}} \log^3 y}{y}, X^{\frac{1}{24}} y^{\frac{5}{4}} \log X \log^7 y}}
+... = \twconst {13} X^{\frac{1}{12}} + 0\parent{\max\parent{\frac{X^{\frac{1}{12}} \log^3 y}{y^{\frac{1}{2}}}, X^{\frac{1}{24}} y^{\frac{5}{4}} \log X \log^7 y}}

-... e^{\frac{1}{6}} ...
+... e^{\frac{1}{4}} ...
```

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```
-... 0\parent{\frac{X^{\frac{1}{24}}}{...}
+... 0\parent{\frac{X^{\frac{1}{12}}}{...}}
```

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```

... &\colonequals \sum_{a \geq 0} \frac{\varphi_{1/2}(p^a)}{1} ...
+... &\colonequals 1 + \sum_{a \geq 1} \frac{\varphi_{1/2}(p^a)}{1} ...

... {13}.
+... {12}.

-      We have been unable to compute  $Q_{13}(13)$  because we do not know  $cT_{13}(13^v, 1)$  and  $cT_{13}(1, 13^v)$  for all  $v \geq 0$ . The square roots appear in \eqref{Equation: Euler factors for m = 13} because of the generalized Jordan totient function  $\varphi_{1/2}$ . For instance, for  $p = 3$  we have
+      \frac{79 \cdot 239 \cdot 719 + 73 \cdot 639 \cdot 763 \sqrt{13}}{79 \cdot 270 \cdot 842}, & \text{if } p=13; \setminus
+      For  $p=13$ , substituting \eqref{Equation: cT13 at 13} into the defining series and summing the geometric tails gives
+      \begin{equation}
+      Q_{13}(13) = \frac{79 \cdot 239 \cdot 719 + 73 \cdot 639 \cdot 763 \sqrt{13}}{79 \cdot 270 \cdot 842} = 4.349 \cdot 034 \cdot 925 \cdot 218 \cdot 983 \cdot 880 \ldots
+      \end{equation}
+      The square roots in \eqref{Equation: Euler factors for m = 13} appear because of the generalized Jordan totient function  $\varphi_{1/2}$ . For instance, for  $p = 3$  we have

... + 7 \sqrt{3}^{36}.
+... + 17 \sqrt{3}^{36}.

```

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```

-0\parent{X^{-1/12} \log X \sum_{n \leq y} 4^{-\omega(n)} ...
+0\parent{X^{-1/24} \log X \sum_{n \leq y} 4^{-\omega(n)} ...

-      We have
-      \begin{equation}
-      \twistNadgy_{13}(X) = 0\parent{\frac{X^{-1/12} \log^3 y}{y^{1/2}}}
-      \end{equation}
-      for  $X, y \geq 2$ .
+For every  $\epsilon > 0$ , we have
+\begin{equation}
+\twistNadgy_{13}(X) = O_{\epsilon} \parent{\frac{X^{-1/12+\epsilon}}{y^{1/6}}} + X^{-1/14+\epsilon}
+\end{equation}
+for  $X, y \geq 2$ .

```

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```

-By \Cref{Lemma: bound on T(e) for m = 13},  $cT_{13}(e_1, e_2) = \dots$  of  $f/n^{\sigma}$ , we conclude \begin{equation} \twistNadgy_m(X) =
0\parent{\frac{X^{-1/12} \log^3 y}{y^{1/2}}} \end{equation} as desired.
+We have  $\text{CII}_{13} = \mathcal{C}7$  and  $\text{CIII}_{13} = \mathcal{C}25$ . The norm-form lattice arguments of \Cref{Lemma: asymptotic for M(X; e) for m = 7,
Lemma: asymptotic for M(X; e) for m = 5 and 10 and 25}, together with the Principle of Lipschitz, give
+\begin{equation}
+\begin{aligned}
+ \# \{(a, b): e^3 \mid \text{CII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E \\
+ \# \{(a, b): e^2 \mid \text{CIII}_{13}(a, b) \} & \text{ for some } e > E
\end{aligned}
+\end{equation}
+for primitive pairs with  $|a|, |b| \leq B$ ; here the exceptional  $3$ - and  $2$ -parts are bounded, since  $v_3(\text{CII}_{13}) \leq 3$  and  $v_2(\text{CIII}_{13}) \leq 3$  for primitive pairs.

+Let  $D = e_{\text{II}} e_{\text{III}} e'$  be the actual twist minimality defect, with  $e' \mid 78$ , as in \Cref{Theorem: Controlling
size of twist minimality defect for m = 13}(b). By \Cref{Proposition: bounding the summands of twN calE X}, the large-defect
sum is bounded, up to  $\tau(D)^2$ , by pairs of twist height at most  $X$  with  $D > y$ . The same theorem gives  $|a|, |b| \ll X^{1/14}$ 
and  $D \ll X^{5/42}$ , so  $\tau(D)^2 \ll \epsilon X$ .

+On a dyadic interval  $D_0 \leq D < 2D_0$ , we have  $|a|, |b| \ll X^{1/24} D_0^{-1/4}$  and either  $e_{\text{II}} \mid D_0^{1/3}$  or  $e_{\text{III}} \mid D_0^{2/3}$ . The preceding estimates therefore give
+\begin{equation}
+O_{\epsilon} \parent{X^{-1/12+\epsilon} D_0^{-1/6} + X^{1/24+\epsilon} D_0^{-1/4}}.
+\end{equation}
+Summing dyadically for  $D_0 > y$  and using  $D_0 \ll X^{5/42}$  proves the lemma.

... 13, which we restate here with an improved error term in the notations we ...
+... 13, which we restate here in the notations we ...

-      We have
+      For every  $\epsilon > 0$ , we have

-      \twistNad_{13}(X) = \twistNad_{13}(X) = \twconst_{13} X^{-1/12} + O\parent{X^{7/108} \log^{43/9} X}
+      \twistNeq_{13}(X) = \twistNad_{13}(X) = \twconst_{13} X^{-1/12} + O_{\epsilon} \parent{X^{4/51+\epsilon}}

-Let  $\delta$  be a positive quantity with  $\delta \ll X^{-1/24}$ . \textit{A fortiori}, we have ... = \twconst_m X^{-1/\deg B} + O\parent{X^{7/108} \log^{43/9} X} \end{equation} as desired.
+Combining \Cref{Lemma: asymptotic for twN <= y(X) for m = 13, Lemma: bound on twN > y(X) for m = 13} and taking  $\delta = X^{-1/34}$ , we have
+\begin{equation}

```

+ $X^{\frac{1}{24}}y^{\frac{5}{4}}=X^{\frac{1}{12}}y^{-\frac{1}{6}}=X^{\frac{4}{51}}$.
+ $\end{equation}$
+The remaining error terms are smaller, and the logarithmic factors are absorbed into X^{ϵ} .

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-... $7/108$
+... $4/51$
- $\mu_{\text{twistLad } \{13\}}(\sigma) < 13/84$
+ $\mu_{\text{twistLad } \{13\}}(\sigma) \leq \max\{0, 17-204\sigma\}$

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-... $7/108$.
+... $4/51$.
- The proof is structurally identical to the one given for $\text{Cref}\{\text{Corollary: twL}(s) \text{ has a meromorphic continuation for } m = 7\}$.
+Abel summation applied to $\text{Cref}\{\text{Theorem: asymptotic for twN}(X) \text{ for } m = 13\}$ gives (a)–(b). On fixed lines $4/51 < \sigma_0 < 1/12$, the remainder integral is $O(1+|t|)$, while absolute convergence gives vertical-growth exponent O for $\sigma > 1/12$; subconvexity of the μ -function and $\text{Cref}\{\text{Theorem: muzeta}(\sigma)\}$ then give (c).
-... analytic behavior of $L_{\text{Qad } m}(s)$ for $m = 10, 25$.
+... analytic behavior of $L_{\text{Qad } \{13\}}(s)$.

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-The sequence $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}$ is admissible ...
+The sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$ is admissible ...
-... admissible for $m = 10$ and 25 , the sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$...
+... admissible for $m = 13$, the sequence $\text{parent}\{\text{hQad } \{13\}(n)\}_{n \geq 1}$...
-... $\{13\}(n)\}_{n \geq 1}$ is admissible with parameters $\text{parent}\{1/6, 1/12, 13/42\}$. We now apply $\text{Cref}\{\text{Theorem: Landau's Tauberian theorem}\}$ to ...
+... $\{13\}(n)\}_{n \geq 1}$ is admissible with parameters $\text{parent}\{1/6, 1/12, 13/84\}$. We now apply $\text{Cref}\{\text{Theorem: Landau's Tauberian theorem}\}$ to ...
-We suspect that the true error on $N_{\text{Qad } m}(X)$ is at most $O(X^{\frac{1}{12} + \epsilon})$, and the true error on $\text{twistNad } \{13\}$...
+We suspect that the true error on $N_{\text{Qad } \{13\}}(X)$ is at most $O(X^{\frac{1}{12} + \epsilon})$, and the true error on $\text{twistNad } \{13\}$...

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-... abiding by its notations, ...
+... using its notation, ...

Submitted Thesis pp. 228–231 (deliberate completion)

-In this section, we describe an algorithm to enumerate the elliptic curves with a cyclic 13 -isogeny and twist height at most X . We then use the list of elliptic curves admitting a cyclic 13 -isogeny to estimate $\text{constad } 13$. However, our ignorance about $\text{cT } \{13\}(13^v, 1)$ and $\text{cT } \{13\}(1, 13^v)$ prevents us from computing $\text{Q } \{13\}(13)$, and thus from computing $\text{twconst } \{13\}$.
+In this section, we describe an algorithm to enumerate the elliptic curves with a cyclic 13 -isogeny and twist height at most X . We then numerically evaluate the Euler product $\text{Q } \{13\}$ and the twist-counting constant $\text{twconst } \{13\}$, and use the list of elliptic curves admitting a cyclic 13 -isogeny to estimate $\text{constad } 13$.
+
+ $\text{subsection}\{\text{Computing } \text{twconst } \{13\}\}$
+
+We first estimate the Euler product $\text{Q } \{13\}$. For $p \notin \{2, 3, 13\}$, set
+ $\begin{equation} A_p \colonequals \frac{2p^{\frac{1}{2}} + (p+1)(p^{\frac{1}{2}}+1)}{(p+1)(p^2+p^{\frac{3}{2}}+p+p^{\frac{1}{2}}+1)}, \\ B_p \colonequals \frac{2p^{\frac{1}{2}} + (p+1)(p^2+p^{\frac{3}{2}}+p+p^{\frac{1}{2}}+1)}{(p+1)(p^2+p^{\frac{3}{2}}+2p^2+p^{\frac{1}{2}}+2)} \end{equation}$
+The first case of $\text{Equation: Euler factors for } m = 13$ satisfies
+ $\begin{equation} \frac{2p^{\frac{1}{2}} + (p+1)(p^2+p^{\frac{3}{2}}+2p^2+p^{\frac{1}{2}}+2)}{(p+1)(p^2+p^{\frac{3}{2}}+2p^2+p^{\frac{1}{2}}+2)} = A_p + B_p \end{equation}$
+Consequently, away from the three exceptional primes, the nontrivial local increment is A_p for $p \equiv 1 \pmod{4}$, together with B_p for $p \equiv 3 \pmod{4}$. In particular,
+ $\begin{equation} 0 < A_p < 2p^{-\frac{3}{2}}, \quad 0 < B_p < 2p^{-\frac{5}{2}}. \end{equation}$
+ $\end{equation}$

+Let $Q_{\{13,\leq Y\}}$ denote the Euler product truncated at $p\leq Y$. Since $\log(1+x)<x$, for $Y\geq 8\cdot 10^9$ the same Abel-summation argument using the explicit prime-counting estimates of Bennett--Martin--O'Bryant--Rechnitzer \cite[Theorem 1.4]{Bennett-Martin-OBryant-Rechnitzer} gives

$$\log\frac{Q_{\{13,\leq Y\}}}{Y^{-1/2}\log Y}\leq \frac{1}{4}\log Y + \frac{1}{4}\log Y + \frac{1}{4}\log Y + \frac{1}{4}\log Y.$$

+At $Y=8\cdot 10^9$, direct computation gives

$$Q_{\{13,\leq Y\}}=9.775\,286\,929\,620\,052\,484\ldots$$

+while the exponent in the preceding upper bound is less than $1.141\,882\cdot 10^{-6}$. Thus

$$9.775\,286\,929\,620\,052 < Q_{\{13\}} < 9.775\,298\,091\,844\,354.$$

+Using the rejection-sampling estimate for $R_{\{13\}}$ below and the finite Euler product as a point estimate, we obtain

$$\text{twconst}_{\{13\}}=\frac{Q_{\{13\}}R_{\{13\}}}{\zeta(2)}\approx 0.968\,679\,323.$$

+The Monte Carlo standard error propagated from $R_{\{13\}}$ is approximately $3.58\cdot 10^{-6}$, while the deterministic uncertainty from truncating the Euler product is less than $1.11\cdot 10^{-6}$. We report these separately, since the former is a statistical standard error rather than an absolute error bound.

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... $b > 0$, we ...
+... $b > 0$ and $A_{\{13\}}(a,b)^{3+27}B_{\{13\}}(a,b)^2 \neq 0$, we ...

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... $2\,\text{twistNad}_{\{13\}}(1/6)/\zeta(2)$...
+... $2\,\text{twistLad}_{\{13\}}(1/6)/\zeta(2)$...

... $0.680\,532\,123\,1018\,161$.
+... $0.680\,532\,123\,101\,821\,172$.

-Ironically, although we have an estimate for $\text{constad}_{\{13\}}$, we are unable to estimate ... $\colonequals 33\,570\,382\,383$ of our first $s_{\{13\}} \colonequals 61\,749\,000\,000$ samples lie in $R_{\{13\}}$, so

+pagebreak

+We estimate $R_{\{13\}}$ by rejection sampling on $[-0.8228, 0.8228] \times [0, 0.1822]$. Of $s_{\{13\}} \colonequals 61\,749\,000\,000$ samples, $r_{\{13\}} \colonequals 33\,570\,382\,383$ lie in $R_{\{13\}}$, giving

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... $8, 9, 12, 16, 18$ in $4, 6, 8, 9, 12, 16, 18$ \label{Chapter: ...}
+... $8, 9, 12, 16, 18$ in $6, 8, 9, 12, 16, 18$ \label{Chapter: ...}

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... $12, 15, \dots$
+... $12, 16, \dots$

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... $\mid g_m(t)$...
+... $\mid f_m(t)$...

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-\$16\$ & 3^2 & 3^5 & 3^8 & 3^{11} & --- \\
+\$16\$ & --- & --- & --- & --- & --- \\

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```
-... b)$ may ...
+... b), \B m (a, b))$ may be ...

-... = $ ...
+... = 4$ ...
```

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```
-\begin{lemma}\label{Lemma: fundamental sieve for for m ...
+\begin{lemma}\label{Lemma: fundamental sieve for m ...

-\twNad 4 (X) = ...
+\twNeq 4 (X) = ...

-... 3} \sum_{e \mid n} \mu(n/e) \cM m (e^6 X; n).
+... 3} \sum_{e \mid n} \mu(n/e) \cM 4 (e^6 X; n).
```

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```
-... m(X; e) = \sum_{\substack{d \mid 11 X^{1/2} \deg B m} \setminus \gcd(d, e) = ...
+... m(X; e) = \sum_{\substack{d \mid 11 X^{1/(2 \deg B m)} \setminus \gcd(d, e) = ...

-... X^{1/2 \deg B m} \{d\}
+... X^{1/(2 \deg B m)} \{d\}

-... 0 \parent{2^{\omega(e)} X^{1/2 \deg B m}} ...
+... 0 \parent{2^{\omega(e)} X^{1/(2 \deg B m)}} ...

- for $X \geq 2$ and $d, e \geq 1$.
+ for $X \geq 2$ and $e \geq 1$.

-... however, because \Cref{Proposition: bounding} ...
+... however, because by \Cref{Proposition: bounding} ...
```

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```
-... 3^4} ...
+... 3^5} ...
```

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```
-... \Cref{Lemma: fundamental sieve for for m of genus $0$} ...
+... \Cref{Lemma: fundamental sieve for m of genus $0$} ...

-... \twconsteq m X^{1/\deg B m} + 0 \parent{X^{1/2 \deg B m} \log X},
+... \twconsteq m X^{1/\deg B m} + 0 \parent{X^{1/(2 \deg B m)} \log X},

-... \Cref{Lemma: fundamental sieve for for m of genus $0$}, ...
+... \Cref{Lemma: fundamental sieve for m of genus $0$}, ...
```

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```
-... 0 \parent{2^{\omega(n)} e^{3/\deg B m} X^{1/2 \deg B m} \log e^6 X}.
+... 0 \parent{2^{\omega(n)} e^{3/\deg B m} X^{1/(2 \deg B m)} \log e^6 X}.

-... and \eqref{Equation: twconst} for ...
+... and \eqref{Equation: twconsteq} for ...
```

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```
- \sum_{n \mid 2^4 \cdot 3^5} \sum_{e \mid n} \mu \parent{n/e} 0 \parent{2^{\omega(n)} e^{3/\deg B m} X^{1/2 \deg B m} \log e^6 X} \setminus
- =& 0 \parent{X^{1/2 \deg B m} \log X \sum_{n \mid 2^4 \cdot 3^5} 2^{\omega(n)} \sum_{e \mid n} \abs{\mu \parent{n/e}} e^{3/\deg B m}} \setminus
- =& 0 \parent{X^{1/2 \deg B m} \log X},
+ \sum_{n \mid 2^4 \cdot 3^5} \sum_{e \mid n} \mu \parent{n/e} 0 \parent{2^{\omega(n)} e^{3/\deg B m} X^{1/(2 \deg B m)} \log e^6 X} \setminus
+ =& 0 \parent{X^{1/(2 \deg B m)} \log X \sum_{n \mid 2^4 \cdot 3^5} 2^{\omega(n)} \sum_{e \mid n} \abs{\mu \parent{n/e}} e^{3/\deg B m}} \setminus
+ =& 0 \parent{X^{1/(2 \deg B m)} \log X},
```

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```
-... asymptotics ...
+... asymptotics ...

-... of \Cref{Theorem: asymptotic for NQ(X)} for  $m = 10$  ...
+... of \Cref{Theorem: asymptotic for  $\text{twN}(X)$ } for  $m = 10$  ...

-... obtain are the the same ...
+... obtain are the same ...

- We record the order of growth for these modular curves, though not the full asymptotics, in Table \ref{table:modularcurves}.
```

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```
-... curfves ...
+... curves ...

- \twistNad  $m(X)$ , \twconstad  $m X^{\{1/\deg B\} m} + O(\text{parent}\{X^{\{1/2\} \deg B\} \log X\})$ ,
+ \twistNad  $m(X) = \text{twconstad } m X^{\{1/\deg B\} m} + O(\text{parent}\{X^{\{1/(2 \deg B)\} \log X\})$ ,

-... = 9, 18$.
+... \in \set{6, 9, 18}$.

-... 6, 8, 9, 10, 12, 16, 18}$. ...
+... 6, 8, 9, 12, 16, 18}$. ...

-...  $1/2 \deg B$ }.
+...  $1/(2 \deg B)$ }.
```

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```
- \mu_{\twistLad  $m$ }(\sigma) < 13/84
+ \mu_{\twistLad  $m$ }(\sigma) \leq \max\{0, 2-2\deg B\} m, \sigma\}

- for  $\sigma > 1/2 \deg B$  m$.
+ for  $\sigma > 1/(2 \deg B) m$ $. The same bound holds for  $\text{twistLeq } m(s)$ $.

-... \set{6, 8, 9, 12, 16, 18}\{m in 4, 6, 8, 9, 12, 16, 18\}\label{Section: ...
+... \set{6, 8, 9, 12, 16, 18}\{m in 6, 8, 9, 12, 16, 18\}\label{Section: ...

-... quadatric ...
+... quadratic ...
```

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```
- Let  $m \in \set{6, 8, 9}$ $. The sequences
+ Let  $m \in \set{6, 8, 9}$ $. The sequences

- are admissible \textup{(\Cref{Definition: admissible sequences})} with parameters  $(1/6, 1/12, 13/42)$ $.
+ are admissible \textup{(\Cref{Definition: admissible sequences})} with parameters  $(\text{parent}\{1/6, \delta, 97\delta/7\})$ $.

- The proof is structurally identical to the one given for \Cref{Lemma: Delta NQ(n)} is admissible for  $m = 7$ }; however, we must run
  through the argument given there twice, once for  $\text{LQeq } m(s)$  and once for  $\text{LQad } m(s)$ $.
+ The proof is structurally identical to the corrected proof of \Cref{Lemma: Delta NQ(n)} is admissible for  $m = 7$ }; however, we
  must run through the argument twice, once for  $\text{LQeq } m(s)$  and once for  $\text{LQad } m(s)$ $.


```

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```
-... \consteqprm  $m X^{\{1/6\} + O(X^{\{1/8 + \epsilon\})}$ 
+... \consteqprm  $m X^{\{1/6\} + O(X^{\{38/291 + \epsilon\})}$ 

-... \constadprm  $m X^{\{1/6\} + O(X^{\{1/8 + \epsilon\})}$ 
+... \constadprm  $m X^{\{1/6\} + O(X^{\{38/291 + \epsilon\})}$ 

- By \Cref{Lemma: Delta NQ(n)} is admissible for  $5 < m \leq 9$ , both  $\text{parent}\{\text{hQeq } m(n)\}_{n \geq 1}$  and  $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}$  are admissible with parameters  $(\text{parent}\{1/6, 1/12, 13/42\})$ $. We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series  $\text{LQeq } m(s)$  and  $\text{LQad } m(s)$ , and our claim follows.
+ Fix  $\epsilon > 0$ $. By \Cref{Lemma: Delta NQ(n)} is admissible for  $5 < m \leq 9$ , for every  $0 < \delta < 7/97$  both  $\text{parent}\{\text{hQeq } m(n)\}_{n \geq 1}$  and  $\text{parent}\{\text{hQad } m(n)\}_{n \geq 1}$  are admissible with parameters  $(\text{parent}\{1/6, \delta, 97\delta/7\})$ $. Since  $97\delta/7 < 1$ $, applying \Cref{Theorem: Landau's Tauberian theorem} and then taking  $\delta$  sufficiently close to  $7/97$  gives the error  $O(X^{\{38/291 + \epsilon\}})$  in both cases.
```

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```
-... m X^{1/6} \log X + O(X^{1/8} ...)
+... m X^{1/6} + O(X^{1/8} ...)

-      By \Cref{Lemma: Delta NQ(n)} is admissible for  $m > 9$ , the sequence  $\left(\sum_{n \geq 1} m(n)\right)$  is admissible with
parameters  $\left(\frac{1}{6}, \frac{1}{12}, \frac{13}{84}\right)$ . We now apply \Cref{Theorem: Landau's Tauberian theorem} to the Dirichlet series  $\sum m(s)$ , and our claim follows.
+      By \Cref{Lemma: Delta NQ(n)} is admissible for  $m > 9$ , both  $\left(\sum_{n \geq 1} m(n)\right)$  and  $\left(\sum_{n \geq 1} m(n)\right)$  are admissible with parameters  $\left(\frac{1}{6}, \frac{1}{12}, \frac{13}{84}\right)$ . We now apply \Cref{Theorem: Landau's Tauberian theorem} to the
Dirichlet series  $\sum m(s)$  and  $\sum m(s)$ , and our claim follows.

-... \set{6, 8, 9, 12, 16, 18}\{m in 4, 6, 8, 9, 12, 16, 18\}
+... \set{6, 8, 9, 12, 16, 18}\{m in 6, 8, 9, 12, 16, 18\}

-\calR m (X) = \calR m (1) X^{1/2 \degB m};
+\calR m (X) = \calR m (1) X^{1/(2 \degB m)};
```

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```
-... X^{1/2 \degB m},
+... X^{1/(2 \degB m)},

-... well as how approximately how ...
+... well as approximately how ...
```

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```
-... $65\,68\,1836\,724$ ...
+... $65\,681\,836\,724$ ...
```

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```
-... error, is estimated based on the sampling recorded in Table \ref{table:Rm}.
+... error from the rejection sampling recorded in Table \ref{table:Rm}, is estimated there as well.
```

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```
-... an error term, ...
+... its propagated standard error, ...

-... Error on  $\mathcal{R} m$  ...
+... Standard error ...

-$16$ & $1$ & $4/3$ ...
+$16$ & $2$ & $4/3$ ...

-... $18$ & $2$ & $(1 ...
+... $18$ & $1$ & $(1 ...

-... Error on  $\text{twconsteq } m$  ...
+... Standard error ...

-...  $\text{twconsteq } m$ , its error, and ...
+...  $\text{twconsteq } m$ , its standard error, and ...

-... Error on  $\text{twconstad } m$  ...
+... Standard error ...
```

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```
-...  $\text{twconstad } m$ , its error, and ...
+...  $\text{twconstad } m$ , its standard error, and ...

-$m$ &  $\text{consteq } m$  & Error on  $\text{consteq } m$  &  $\text{constad } m$  & Error on  $\text{constad } m$  \\
+$m$ &  $\text{consteq } m$  & Standard error &  $\text{constad } m$  & Standard error \\
```

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```

-... 3 \zeta(2)$ ...
+... (3 \zeta(2))$ ...

-... / 3 \zeta(2)$, whence ...
+... / (3 \zeta(2))$, whence ...

-$6$ & $0.153\,033\,271$ & $4.8 \cdot 10^{-7}$ & $0.153\,033\,271$ & $4.8 \cdot 10^{-7}$ \\
-$8$ & $0.118\,663\,783$ & $4.4 \cdot 10^{-7}$ & $0.059\,331\,892$ & $2.2 \cdot 10^{-7}$ \\
-$9$ & $0.094\,668\,211$ & $1.7 \cdot 10^{-7}$ & $0.094\,668\,211$ & $1.7 \cdot 10^{-7}$ \\
-\end{tabular}}{The constants $\consteq$ m$ and $\constad$ m$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $0.153\,033\,271$ & $6.1 \cdot 10^{-7}$ & $0.153\,033\,271$ & $6.1 \cdot 10^{-7}$ \\
+$8$ & $0.118\,663\,783$ & $5.5 \cdot 10^{-7}$ & $0.059\,331\,892$ & $2.9 \cdot 10^{-7}$ \\
+$9$ & $0.094\,668\,211$ & $2.1 \cdot 10^{-7}$ & $0.094\,668\,211$ & $2.1 \cdot 10^{-7}$ \\
+\end{tabular}}{The constants $\consteq$ m$ and $\constad$ m$ and their standard errors for $m$ in $\set{6, 8, 9}$}

-&\abs{\int_X^{\infty} \parent{\twistNad 7(u) - \twconst 7 \floor{u^{1/6}}} u^{-7/6} \, , \mathrm{d}u} \\
&\abs{\int_X^{\infty} \parent{\twistNad m(u) - \twconst m \floor{u^{1/6}}} u^{-7/6} \, , \mathrm{d}u} \\

-We do not know the exact value for $M$, but empirically, we find that for $1 \leq u \leq 10^{42}$, we have
+We do not know the exact value for $M$, but empirically, we find that throughout the range $10^{19} \leq u \leq 10^{42}$, we have

```

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```

-If we assume these bounds continue to hold for larger $u$, and combine this ... \ref{table:ell}, and the estimates for $\consteqpm$
m$ and $\constadpm$ m$ given in Table \ref{table:constm}.
+If we assume these bounds continue to hold for larger $u$, and combine the resulting truncation error with the standard errors in
our approximations for $\twconsteq$ m$ and $\twconstad$ m$, we obtain the uncertainties in Table \ref{table:ell} and Table \ref{
table:constm}.

-$m$ & $\widetilde{ell}_{m, 0}$ & Error on $\widetilde{ell}_{m, 0}$ & $\ell_{m, 0}$ & Error on $\ell_{m, 0}$ \\
+$m$ & $\widetilde{ell}_{m, 0}$ & Uncertainty & $\ell_{m, 0}$ & Uncertainty \\

-$6$ & $-0.572\,182\,786$ & $1.13 \cdot 10^{-1}$ & $-0.636\,153\,135$ & $1.15 \cdot 10^{-1}$ \\ ... $\widetilde{ell}_{m, 0}$ and $\ell_{m, 0}$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $-0.572\,182\,786$ & $1.21 \cdot 10^{-2}$ & $-0.677\,298\,843$ & $1.21 \cdot 10^{-2}$ \\
+$8$ & $-0.742\,599\,375$ & $9.33 \cdot 10^{-3}$ & $-0.387\,595\,153$ & $4.65 \cdot 10^{-3}$ \\
+$9$ & $-0.231\,206\,039$ & $9.37 \cdot 10^{-3}$ & $-0.231\,206\,039$ & $9.37 \cdot 10^{-3}$ \\
+\end{tabular}}{The constants $\widetilde{ell}_{m, 0}$ and $\ell_{m, 0}$ and their numerical uncertainties for $m$ in $\set{6, 8, 9}$}
}

-$m$ & $\consteqpm$ m$ & Error on $\consteqpm$ m$ & $\constadpm$ m$ & Error on $\constadpm$ m$ \\
+$m$ & $\consteqpm$ m$ & Uncertainty & $\constadpm$ m$ & Uncertainty \\

-$6$ & $-0.037\,215\,321$ & $1.78 \cdot 10^{-1}$ & $-0.114\,993\,938$ & $1.78 \cdot 10^{-1}$ \\ ... $\consteqpm$ m$ and $\constadpm$
m$ and their error for $m$ in $\set{6, 8, 9}$}
+$6$ & $-0.037\,215\,321$ & $1.47 \cdot 10^{-2}$ & $-0.165\,021\,121$ & $1.47 \cdot 10^{-2}$ \\
+$8$ & $-0.392\,302\,971$ & $1.14 \cdot 10^{-2}$ & $-0.215\,964\,396$ & $5.65 \cdot 10^{-3}$ \\
+$9$ & $0.126\,227\,997$ & $1.14 \cdot 10^{-2}$ & $0.126\,227\,997$ & $1.14 \cdot 10^{-2}$ \\
+\end{tabular}}{The constants $\consteqpm$ m$ and $\constadpm$ m$ and their numerical uncertainties for $m$ in $\set{6, 8, 9}$}

```

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```

-\frac{\twistNeq m(X)}{X^{1/6}} ...
+\frac{N\!Qeq m(X)}{X^{1/6}} ...

-\frac{\twistNad m(X)}{X^{1/6}} ...
+\frac{N\!Qad m(X)}{X^{1/6}} ...

-... $\frac{\twistNeq}{...}$
+... $\frac{N\!Qeq}{...}$

-... $\frac{\twistNad}{...}$
+... $\frac{N\!Qad}{...}$

-... $146\,844\,192$ & $-0.115\,210\,235$ ...
+... $146\,343\,642$ & $-0.165\,265\,235$ ...

-... $55\,418\,512$ & $-0.196\,051\,423$ ...
+... $55\,219\,676$ & $-0.215\,935\,023$ ...

```

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```

-... 2 \twistNeq ...
+... 2 \twistLeq ...

-... 2 \twistNad {m}(1/6)/\zeta(2),
+... 2 \twistLad {m}(1/6)/\zeta(2),

```

```

-... 0.212\,842\,775\,719\,189\,16, ...
+... 0.212\,842\,775\,719\,150\,50, ...

-... 0.106\,421\,387\,859\,584\,87, ...
+... 0.106\,421\,387\,859\,575\,25, ...

-... 0.269\,169\,745\,679\,629\,80, ...
+... 0.269\,169\,745\,679\,529\,87, ...

-... 0.134\,584\,872\,839\,728\,06, ...
+... 0.134\,584\,872\,839\,764\,93, ...

-... 0.107\,025\,809\,031\,522\,12.
+... 0.107\,025\,809\,032\,944\,72.

```

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```

-If these bounds continue to hold for larger  $X$ , then
+If these bounds continue to hold for larger  $X$ , then partial summation, using the endpoint values from Table \ref{table:enumerating
}, gives

-\sum_{n > 10^{84}} \frac{\twistheq{12}(n)}{n^{1/6}} \approx \int_{10^{84}}^{\infty} x^{-1/6} d \twistNeq{12}(x) < 1.090\,472 X^{1/12} ...
10^{126}} \frac{\twisthad{18}(n)}{n^{1/6}} \approx \int_{10^{126}}^{\infty} x^{-1/6} d \twistNad{18}(x) < 5.035\,48 \cdot 10^{-15}.
\\
+\sum_{n > 10^{84}} \frac{\twistheq{12}(n)}{n^{1/6}} < 1.133\,046\,8 \cdot 10^{-7}, \\
+\sum_{n > 10^{84}} \frac{\twisthad{12}(n)}{n^{1/6}} < 5.665\,234 \cdot 10^{-8}, \\
+\sum_{n > 10^{84}} \frac{\twistheq{16}(n)}{n^{1/6}} < 9.118\,462 \cdot 10^{-8}, \\
+\sum_{n > 10^{84}} \frac{\twisthad{16}(n)}{n^{1/6}} < 4.559\,231 \cdot 10^{-8}, \text{and} \\
+\sum_{n > 10^{126}} \frac{\twisthad{18}(n)}{n^{1/6}} < 5.375\,8 \cdot 10^{-15}.

-... estimates ...
+... estimates ...

-$m$ & $\consteq m$ & Error on $\consteq m$ & $\constad m$ & Error on $\constad m$ \\
+$m$ & $\consteq m$ & Error bound & $\constad m$ & Error bound \\

-$12$ & $0.258\,785\,783$ & $1.4 \cdot 10^{-7}$ & $0.129\,392\,891$ & $6.7 \cdot 10^{-8}$ \\
their error for $m \in \{12, 16, 18\}$ \\
+$12$ & $0.258\,785\,784$ & $1.4 \cdot 10^{-7}$ & $0.129\,392\,892$ & $6.9 \cdot 10^{-8}$ \\
+$16$ & $0.327\,271\,167$ & $1.2 \cdot 10^{-7}$ & $0.163\,635\,583$ & $5.6 \cdot 10^{-8}$ \\
+$18$ & $0.130\,127\,779\,817\,961\,2$ & $6.6 \cdot 10^{-15}$ & $0.130\,127\,779\,817\,961\,2$ & $6.6 \cdot 10^{-15}$ \\
+\end{tabular}}{The constants $\consteq m$ and $\constad m$ and their conditional error bounds for $m \in \set{12, 16, 18}$}

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-In each of these cases, the compactified moduli space  $X_0(m)$  is of genus  $g > 0$ , so by Faltings's theorem (\Cref{Theorem:
Faltings Theorem}), each curve has a finite number of rational points.
+In each of these cases, the compactified moduli space  $X_0(m)$  has genus  $g > 0$  and only finitely many rational points.

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-... curves admitting cyclic  $m$ -isogeny, ...
+... curves admitting a cyclic  $m$ -isogeny, ...

-... 37, 43, 163}.
+... 37, 43, 67, 163}.

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-The following table is extracted from \cite[page 1]{Mazur2}.
+The twist classes and their  $j$ -invariants are classical; see \cite[page 1]{Mazur2} and the references there. We use twist-minimal
short Weierstrass representatives with  $B > 0$ .

-...  $(-1\,8075, \dots$ 
+...  $(-18\,075, \dots$ 

-$27$ &  $(-270, -1\,708)$  &  $78\,766\,128$  &  $-12288000$  \\
+$27$ &  $(-120, 506)$  &  $6\,912\,972$  &  $-12288000$  \\

-...  $730\,6267\,500$  ...
+...  $7\,306\,267\,500$  ...

-...  $6\,313\,1603\,150$  ...
+...  $63\,131\,603\,150$  ...

-$43$ &  $(-3\,440, 42)$  &  $162\,830\,336\,000$  &  $-884736000$ 

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+\$43\$ & \$(-3\,440, 77\,658)\$ & \$162\,830\,654\,028\$ & \$-884736000\$ \\\

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-By the Mazur's theorem ...
+By Mazur's theorem ...

-... Table \ref{table:nonzerogenus} it is an ...
+... Table \ref{table:nonzerogenus} is an ...

-... curves admitting cyclic m -isogeny ...
+... curves admitting a cyclic m -isogeny ...

-... 2, 3, 4, 5, 6, 8, 9, 10, 12, ...
+... 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, ...

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-... 78\,766\,128), ...
+... 6\,912\,972), ...

-... 730\,6267\,500) ...
+... 7\,306\,267\,500) ...

-... 162\,830\,336\,000), ...
+... 162\,830\,654\,028), ...

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-... additional ...
+... additional ...

-... 37, 43, 163},
+... 37, 43, 67, 163},

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-\N{eq {11}} (X) = & \kfree 2\parent{\parent{X/77\,480\,172}^{1/6}} + \kfree 2\parent{\parent{X/2\,923\,690\,572}^{1/6}} \\\
-&+ \kfree 2\parent{\parent{X/323\,181\,166\,572}^{1/6}}, \\\
+\N{eq {11}} (X) = & 2 \kfree 2\parent{\parent{X/77\,480\,172}^{1/6}} + 2 \kfree 2\parent{\parent{X/2\,923\,690\,572}^{1/6}} \\\
+&+ 2 \kfree 2\parent{\parent{X/323\,181\,166\,572}^{1/6}}, \\\

-\N{eq {14}} (X) = & \kfree 2\parent{\parent{X/259\,308}^{1/6}} + \kfree 2\parent{\parent{X/842\,579\,500}^{1/6}}, \\\
+\N{eq {14}} (X) = & 2 \kfree 2\parent{\parent{X/259\,308}^{1/6}} + 2 \kfree 2\parent{\parent{X/842\,579\,500}^{1/6}}, \\\

-\N{eq {15}} (X) = & \kfree 2\parent{\parent{X/234\,967\,500}^{1/6}} + \kfree 2\parent{\parent{X/478\,550\,700}^{1/6}} \\\
-&+ \kfree 2\parent{\parent{X/126\,818\,068\,500}^{1/6}} + \kfree 2\parent{\parent{X/23\,621\,749\,807\,500}^{1/6}}, \\\
+\N{eq {15}} (X) = & 2 \kfree 2\parent{\parent{X/234\,967\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/478\,550\,700}^{1/6}} \\\
+&+ 2 \kfree 2\parent{\parent{X/126\,818\,068\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/23\,621\,749\,807\,500}^{1/6}}, \\\

-\N{eq {17}} (X) = & \kfree 2\parent{\parent{X/4\,325\,629\,743\,607\,500}^{1/6}} + \kfree 2\parent{\parent{X/335\,734\,876\,712\,407\,500}^{1/6}}, \\\
+\N{eq {17}} (X) = & 2 \kfree 2\parent{\parent{X/4\,325\,629\,743\,607\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/335\,734\,876\,712\,407\,500}^{1/6}}, \\\

-... (X) = & \kfree 2\parent{\parent{X/14\,074\,668}^{1/6}}, ...
+... (X) = & 2 \kfree 2\parent{\parent{X/14\,074\,668}^{1/6}}, ...

-\N{eq {21}} (X) = & \kfree 2\parent{\parent{X/364\,500}^{1/6}} + \kfree 2\parent{\parent{X/1\,853\,388}^{1/6}} \\\
-&+ \kfree 2\parent{\parent{X/57\,395\,607\,372}^{1/6}} + \kfree 2\parent{\parent{X/20\,478\,321\,699\,372}^{1/6}}, \\\
+\N{eq {21}} (X) = & 2 \kfree 2\parent{\parent{X/364\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/1\,853\,388}^{1/6}} \\\
+&+ 2 \kfree 2\parent{\parent{X/57\,395\,607\,372}^{1/6}} + 2 \kfree 2\parent{\parent{X/20\,478\,321\,699\,372}^{1/6}}, \\\

-\N{eq {27}} (X) = & \kfree 2\parent{\parent{X/78\,766\,128}^{1/6}}, \\\
+\N{eq {27}} (X) = & 2 \kfree 2\parent{\parent{X/6\,912\,972}^{1/6}}, \\\

-\N{eq {37}} (X) = & \kfree 2\parent{\parent{X/730\,6267\,500}^{1/6}} + \kfree 2\parent{\parent{X/107\,611\,181\,539\,805\,427\,907\,500}^{1/6}}, \\\
+\N{eq {37}} (X) = & 2 \kfree 2\parent{\parent{X/7\,306\,267\,500}^{1/6}} + 2 \kfree 2\parent{\parent{X/107\,611\,181\,539\,805\,427\,907\,500}^{1/6}}, \\\

-\N{eq {43}} (X) = & \kfree 2\parent{\parent{X/162\,830\,336\,000}^{1/6}}, \\\
+\N{eq {43}} (X) = & 2 \kfree 2\parent{\parent{X/162\,830\,654\,028}^{1/6}}, \\\

-... (X) = & \kfree 2\parent{\parent{X/102\,480\,782\,771\,052}^{1/6}}, ...
+... (X) = & 2 \kfree 2\parent{\parent{X/102\,480\,782\,771\,052}^{1/6}}, ...

```

-... (X) =& \kfree 2\parent{\parent{X/2\,631\,905\,352\,272\,628\,650\,988}~{1/6}}.
+... (X) =& 2 \kfree 2\parent{\parent{X/2\,631\,905\,352\,272\,628\,650\,988}~{1/6}}.

-=& \kfree 2 ...
+=& 2 \kfree 2 ...

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-... the present ...
+... the presence ...

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-$11$ & $0.05285852537804229\ldots$ \ $14$ & $0.09590984282353528\ldots$ \ $15$ & $0.05837531634681239\ldots$ \ $17$ & ... $0
.013888883070281625\ldots$ \ $43$ & $0.00822676234970696\ldots$ \ $67$ & $0.002810246610438085\ldots$ \ $163$ & $0
.00016360872509265466\ldots$ \
+$11$ & $0.10571705075608457\ldots$ \
+$14$ & $0.19181968564707050\ldots$ \
+$15$ & $0.11675063269362475\ldots$ \
+$17$ & $0.0044705452369290256\ldots$ \
+$19$ & $0.07824834140601367\ldots$ \
+$21$ & $0.28048805576004344\ldots$ \
+$27$ & $0.08809218206211060\ldots$ \
+$37$ & $0.027777766140563242\ldots$ \
+$43$ & $0.01645351934345495\ldots$ \
+$67$ & $0.005620493220876168\ldots$ \
+$163$ & $0.00032721745018530919\ldots$ \

```